1 Vector Analysis and Electrostatics-I

VECTOR ANALYSIS

Electromagnetic theory deals with the study of charges at rest and in motion. Electromagnetic principles are fundamental to the study of electrical / electronics engineering and physics. In this, the quantities involved can be categorized as source quantities and field quantities. Electromagnetic theory deals directly with the electric field and magnetic field vectors. Vector algebra is a mathematical tool with which electromagnetic concepts are more conveniently expressed and best comprehended.

VECTOR ALGEBRA

There are three types of physical quantities in vector algebra which are based on magnitude and direction.

Scalar Quantity:

(I) A quantity which is characterized by only its magnitude and algebraic sign. It does not have any association with direction.

(II) Symbols for scalar quantities are usually represented by any English letter. Examples are Potential → V and Energy → E.

(III) Quantities such as time, mass, distance, temperature, electric potential are the scalar quantities.

Vector Quantity:

(I) A quantity which has magnitude, direction and also obey vector law of addition.

(II) Symbols for vector quantities are usually represented by placing an arrow above the quantity notation.

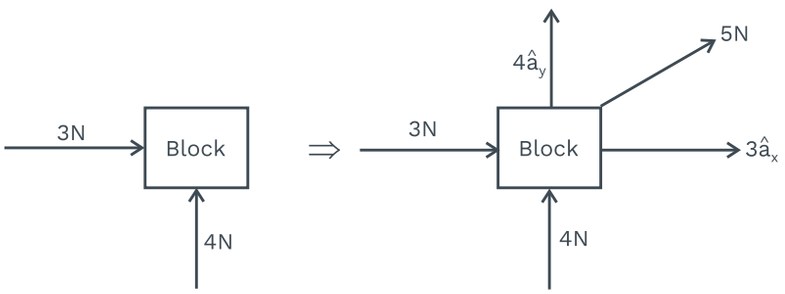
For example: Electric field intensity → $\vec{E}$

Magnetic field Intensity → $\vec{H}$

(III) Vector quantities include velocity, force, displacement, and electric field intensity.

Example 1:

Consider a block on which two forces 3N and 4N are applied as shown below. Calculate the net force acting on the block.



Vector Quantity

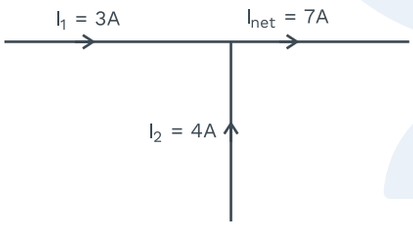
Solution:
The net force acting on the block is:

Vector: $\vec{F} = 3\hat{a}_x + 4\hat{a}_y \text{ N}$

Magnitude: $\left|\vec{F}\right| = \sqrt{(3)^2 + (4)^2} = \sqrt{9 + 16} = 5\text{N}$

Tensor quantity:
A tensor is a quantity which has magnitude, direction, and a plane in which it acts. Stress and strain are both tensor quantities.

Example 2:
Consider two currents of 3A and 4A passing through a junction of wire as shown in the figure. Calculate the net current.



Tensor Quantity

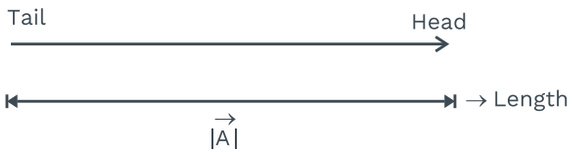
Solution:
The net current is the algebraic sum of two currents.

$$I_{net} = I_1 + I_2 = 3 + 4 = 7\text{A}$$

Representation of Vector:
The mathematical representation of vector is $\vec{A} = \left|\vec{A}\right|\hat{a}_A$, where $\left|\vec{A}\right|$ is the magnitude and $\hat{a}_A$ is the direction of vector $\vec{A}$. Consider a vector $\vec{A}$ with two points tail and head as shown below. The length from tail to head gives the magnitude of vector $\vec{A}$. The head gives the direction of vector.

Definitions

Field:
A field is a function that specifies a particular quantity everywhere in the region.



Vector Representation

Component of Vector:

Consider a vector $\vec{A}$ with tail and head. The length of the vector is unit length.

$$\vec{A} = |\vec{A}|\,\hat{a}_A$$



Component of Vector

Wherever the unit length component of a vector is in the direction of main vector, known as unit vector. Here the unit represent the length of vector $\vec{A}$.

Unit Vector

(1) It is the vector of unit magnitude.
(2) It represents the direction of main vector.
(3) A unit vector is always in the direction, where main vector increases.
(4) Formula for unit vector is

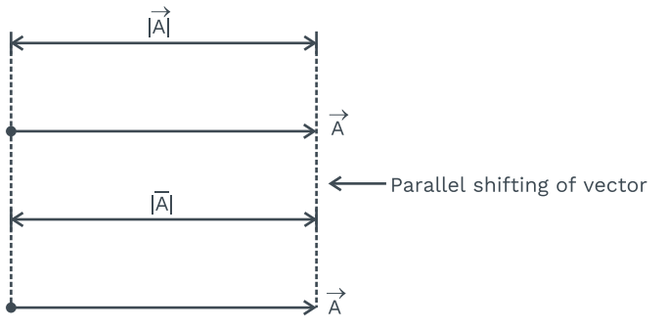
$$\hat{a}_A = \frac{\vec{A}}{|\vec{A}|} = \frac{\text{Main vector}\,(\vec{A})}{\text{Magnitude}\,(\vec{A})}\ .$$

Definitions

A unit vector $\hat{a}_A$ along $\vec{A}$ is defined as a vector whose magnitude is unity and its direction is along $\vec{A}$ which is given by $\hat{a}_A = \dfrac{\vec{A}}{|\vec{A}|}$

Parallel Shifting of Vector:

Consider a vector $\vec{A}$ which is shifted parallelly to some other position. Now, the direction and magnitude of the shifted vector remains unchanged whenever a vector is shifted parallelly.



Parallel Shifting of a Vector

Angle Between Two Vectors:

Consider two vectors $\vec{A}$ and $\vec{B}$, the angle between two vectors must follow below conditions:

(1) To calculate the angle between two vectors, the vectors must be connected tail to tail. If the vectors are not connected from tail to tail, then join them by using parallel shifting of vectors.

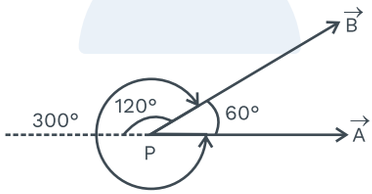
(2) Now, calculate the angle between vector $\vec{A}$ and vector $\vec{B}$.

$0 \leq \theta_{AB} \leq 180°$

(3) The angle θ_{AB} should not be considered less than 0° and greater than 180°.

Example 3:

Calculate the angle between vector $\vec{A}$ and vector $\vec{B}$ from the below figure.



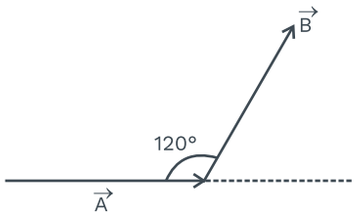
Parallel Shifting of a Vector

Solution:

To calculate angle between two vectors, we need to connect tail to tail of two vectors so, at point P, two vectors are connected in tail to tail then the angle which is between 0° to 180° is the angle θ_{AB}. Therefore, θ_{AB} is 60° it should not be 300° or 120°.

Example 4:

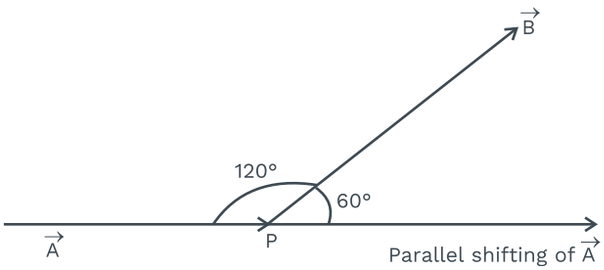
Consider two vectors $\vec{A}$ and $\vec{B}$. Find the angle between $\vec{A}$ and $\vec{B}$ from the figure given below.



Parallel Shifting of a Vector

Solution:

By using parallel shifting of vector, we can find the angle between $\vec{A}$ and $\vec{B}$.



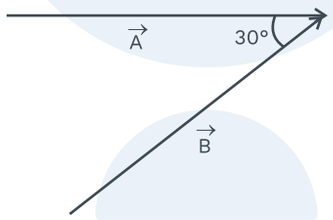
Parallel Shifting of a Vector

Now, the tail-to-tail connection is established, so the angle between $\vec{A}$ and $\vec{B}$ is 60°.

$\therefore \quad \theta_{AB} = 60°$

Example 5:

Calculate the angle θ_{AB} from the given vector diagram.

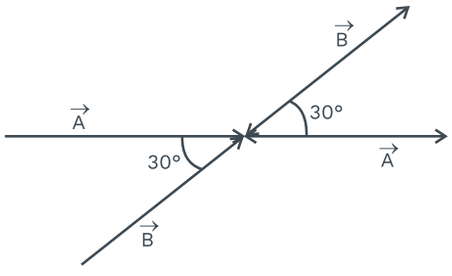


Parallel Shifting of a Vector

Solution:

After applying parallel shifting of vectors, the tail-to-tail connection is established.

Now the angle is $\theta_{AB} = \theta_{BA} = 30°$



Parallel Shifting of a Vector

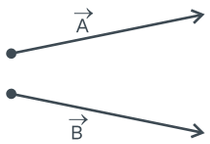
Vector Law of Addition:

Two vectors $\vec{A}$ and $\vec{B}$ can be added together to give another vector $\vec{C}$ by using following two laws:

1. Triangle law of Addition

This law is used to add two vectors, when the first vector's head is joined to the tail of the second vector and then joining the tail of the first vector to the head of the second vector to form a triangle, and hence obtain the resultant sum vector.

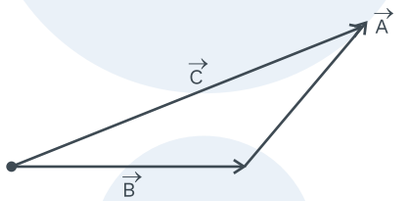
Let's assume two vectors $\vec{A}$ and $\vec{B}$ with their tail and heads as shown below.



Vector Addition

Applying triangle law of addition, then the tail of vector $\vec{A}$ is connected to the head of vector $\vec{B}$. The resultant vector is the sum of two vectors.

$$\vec{C} = \vec{A} + \vec{B}$$

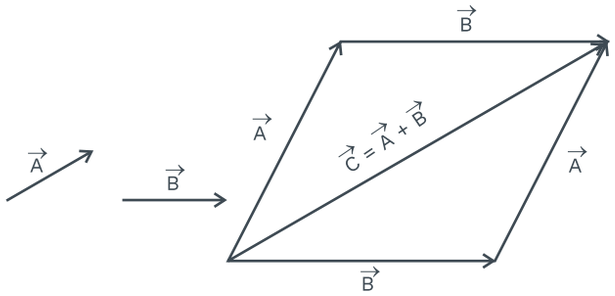


Triangle law of Vector Addition

2. Parallelogram law of Vector Addition:

In this, one tail of a vector is connected to another tail of a vector. Now shift the vector $\vec{A}$ and vector $\vec{B}$ parallel to obtain a parallelogram. From this, larger diagonal $(\vec{C})$ will give the resultant addition.

Let's assume two vectors $\vec{A}$ and $\vec{B}$ with their tail and head. To calculate addition of two vectors using parallelogram law of addition, connect the tail of vector $\vec{A}$ and $\vec{B}$ as shown. The resultant addition is $\vec{C} = \vec{A} + \vec{B}$.



Parallelogram law of Vector Addition

Multiplication of Vectors

If two vectors $\vec{A}$ and $\vec{B}$ are multiplied, the result is either a scalar or a vector depending on how they are multiplied. There are two types of vector multiplication

1) Dot Product.
2) Cross Product.

Scalar Product of Two Vectors (Dot Product)

Let's consider two vectors $\vec{A}$ and $\vec{B}$

$$\vec{A} = |\vec{A}|\hat{a}_A = A\hat{a}_A$$

$$\vec{B} = |\vec{B}|\hat{a}_B = B\hat{a}_B$$

The product is given by $\vec{A}.\vec{B} = |\vec{A}||\vec{B}|\cos\theta_{AB}$ (or) $\vec{A}\cdot\vec{B} = \bar{A}\cdot\bar{B}\cos\theta_{AB}$

Here, $|\vec{A}|$ and $|\vec{B}|$ are magnitudes, θ_{AB} is the angle between $\vec{A}$ and $\vec{B}$ which is obtained by tail to tail connection of $\vec{A}$ and $\vec{B}$ $(0 \le \theta_{AB} \le 180^\circ)$.

Properties of Dot Product:

1. It results in a scalar quantity.

2. Commutative in nature $\Rightarrow \vec{A}.\vec{B} = \vec{B}.\vec{A}$

3. Distributive in nature $\Rightarrow \vec{A}.(\vec{B}+\vec{C}) = \vec{A}.\vec{B}+\vec{A}.\vec{C}$

4. $\vec{A}.\vec{A} = |\vec{A}||\vec{A}|\cos 0^\circ$

$\vec{A}.\vec{A} = |\vec{A}||\vec{A}| = |\vec{A}|^2 = A^2$

5. $\vec{A}.\vec{B} = AB\cos\theta_{AB}$

(i) If $\theta_{AB} = 0°$ then the two vectors are parallel to each other then

$\vec{A}.\vec{B} = A.B$

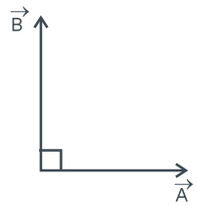


Parallel Vectors

(ii) If $\theta_{AB} = (2n+1)\frac{\pi}{2} = 90^\circ$ then the two vectors are perpendicular to each other.

Definitions

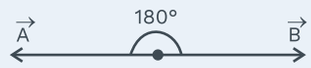
The dot product of two vectors $\vec{A}$ and $\vec{B}$, written as $\vec{A}.\vec{B}$, is defined geometrically as the product of the magnitudes of $\vec{A}$ and $\vec{B}$ and the cosine of the smaller angle between them when they are drawn tail to tail.



Perpendicular Vectors

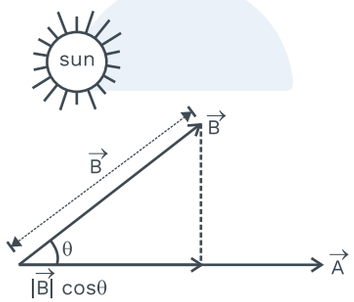
$\vec{A}.\vec{B} = 0$. Where $\vec{A}$ and $\vec{B}$ are perpendicular (or) orthogonal vectors.

(iii) If $\theta_{AB} = 180°$, then the two vectors are connected tail to tail. Then $\vec{A}.\vec{B} = -AB$, where $\vec{A}$ and $\vec{B}$ are anti parallel vectors.



Anti Parallel Vectors

6. The angle cos θ_{AB} lies between (–1 to 1) such that $\left|\vec{A}.\vec{B}\right| \leq \vec{A}.\vec{B}$
7. Physical significance of dot product.



Dot Product Significance

The length of the shadow of $\vec{B}$ = Projection of $\vec{B}$ along vector $\vec{A}$. Here the projection represents the length of vector quantity = B cos θ

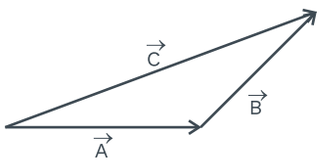
$$\vec{A}.\vec{B} = A(B\cos\theta)$$

$\vec{A}.\vec{B} = A$ [Projection of $\vec{B}$ along vector $\vec{A}$]

Example 6:

A vector $\vec{C} = \vec{A} + \vec{B}$. The magnitude of vector $|\vec{C}|$ is.

Solution:



Vector Product

$$\vec{C} = \vec{A} + \vec{B}$$

$$\vec{C} \cdot \vec{C} = (\vec{A} + \vec{B}) \cdot (\vec{A} + \vec{B})$$

$$|\vec{C}|^2 = \vec{A} \cdot \vec{A} + \vec{A} \cdot \vec{B} + \vec{B} \cdot \vec{A} + \vec{B} \cdot \vec{B}$$

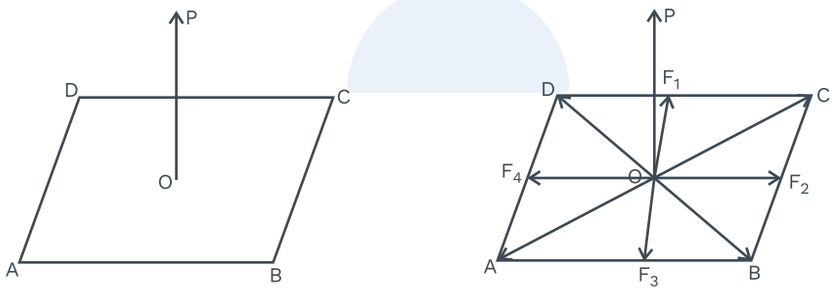
$$C^2 = A^2 + AB\cos\theta + BA\cos\theta + B^2$$

$$C = \sqrt{A^2 + B^2 + 2AB\cos\theta_{AB}}$$

Note:

If any vector is perpendicular to any surface, then that vector will become perpendicular to all the vectors which are drawn on the surface.

$\overline{OP} \perp (ABCD)$ surface, then $\overline{OP} \perp (OF)_n$ where n = 1, 2, 3, 4



Perpendicular Vectors

Cross Product between Two Vectors

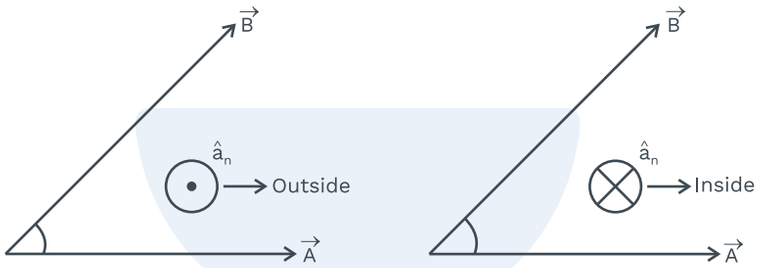
Consider two vectors $\vec{A} = |\vec{A}|\hat{a}_A$ and $\vec{B} = |\vec{B}|\hat{a}_B$. The cross product is given by $\vec{A} \times \vec{B} = |\vec{A}||\vec{B}|\sin\theta_{AB}\hat{a}_n$ or $\vec{A} \times \vec{B} = (AB\sin\theta_{AB})\hat{a}_n$, where $AB\sin\theta_{AB}$ is magnitude and $\hat{a}_n$ is direction. It can also be represented by $\vec{C} = \vec{A} \times \vec{B} = AB\sin\theta_{AB}\hat{a}_n$ or $|\vec{C}| = |\vec{A} \times \vec{B}| = AB\sin\theta_{AB}$.

And also, $\hat{a}_C = \hat{a}_n \rightarrow$ Unit vector of $\vec{A} \times \vec{B}$ or the direction of $\vec{A} \times \vec{B}$

Calculation of $\hat{a}_n$:

The direction of $\vec{A} \times \vec{B}$ can be found by using right-hand rule, place right-hand such that four fingers along vector $\vec{A}$ and rotate fingers towards vector $\vec{B}$. So, that direction of right-hand thumb will give the direction of $\vec{A} \times \vec{B}$ or $\hat{a}_n$. The direction of $\hat{a}_n$ is always be perpendicular to a plane in which both $\vec{A}$ and $\vec{B}$ are present.

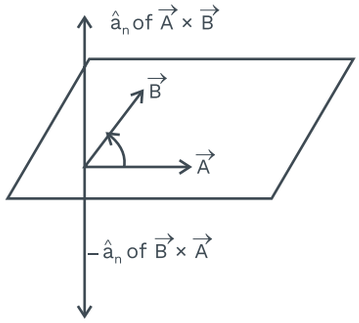
For example:



Direction of a Vector Calculation

The direction of $\vec{A} \times \vec{B}$: $\hat{a}_n = \odot$

and direction of $\vec{B} \times \vec{A}$: $-\hat{a}_n = \otimes$



Direction of a Vector

Properties of Cross Product:

(1) $\vec{C} = \vec{A} \times \vec{B} = AB \sin\theta_{AB} \hat{a}_n$

(2) $\vec{A} \times \vec{B}$ is parallel to $\hat{a}_n$

(3) $\vec{A} \times \vec{B}$ is perpendicular to $\vec{A}$

(4) $\vec{A} \times \vec{B}$ is perpendicular to $\vec{B}$

(5) $\vec{A} \times \vec{B} = -\vec{B} \times \vec{A}$: Not commutative

(6) $\vec{A} \times (\vec{B} + \vec{C}) = \vec{A} \times \vec{B} + \vec{A} \times \vec{C}$: Distributive

(7) If vector $\vec{A}$ is parallel (||) to vector $\vec{B}$, then angle between $\vec{A}$ & $\vec{B} = 0°$

$$\vec{A} \parallel \vec{B} \rightarrow \theta_{AB} = 0°$$

$$\vec{A} \times \vec{B} = AB \sin\theta_{AB} \hat{a}_n = 0$$

Also, if vector $\vec{A}$ is anti-parallel to vector $\vec{B}$. Then the angle between $\vec{A}$ and $\vec{B}$ is 180°.

$$\vec{A} \uparrow\downarrow \vec{B} \rightarrow \theta_{AB} = 180°$$

$$\vec{A} \times \vec{B} = AB \sin\theta_{AB} \hat{a}_n = 0$$

Therefore, If any two vectors are parallel or antiparallel, then their cross product will always be zero.

(8) If vector $\vec{A}$ is perpendicular to vector $\vec{B}$, then the angle between $\vec{A}$ and $\vec{B}$ is 90°. The cross product is $\vec{A} \times \vec{B} = AB\hat{a}_n$

Note:

(1) If $\vec{A} . \vec{B} = AB \cos\theta_{AB}$

$$\cos\theta_{AB} = \frac{\vec{A} . \vec{B}}{AB}$$

$$\theta_{AB} = \cos^{-1}\left[\frac{\vec{A} . \vec{B}}{A.B}\right] \rightarrow \text{Equation (1)}$$

(2) If $\vec{A} \times \vec{B} = AB \sin\theta_{AB} \hat{a}_n$

$$|\vec{A} \times \vec{B}| = AB \sin\theta_{AB}$$

$$\theta_{AB} = \sin^{-1}\left[\frac{|\vec{A} \times \vec{B}|}{A.B}\right] \rightarrow \text{Equation (2)}$$

From equation (1) and (2), equation (2) will give the more appropriate value.

Scalar Triple Product:

Consider three vectors $\vec{A}, \vec{B}$ and $\vec{C}$, where

$$\vec{A} = A_1\hat{i} + A_2\hat{j} + A_3\hat{k}$$

$$\vec{B} = B_1\hat{i} + B_2\hat{j} + B_3\hat{k}$$

$$\vec{C} = C_1\hat{i} + C_2\hat{j} + C_3\hat{k}$$

The scalar product is given by

$$\vec{A}.(\vec{B}\times\vec{C}) = \begin{vmatrix} A_1 & A_2 & A_3 \\ B_1 & B_2 & B_3 \\ C_1 & C_2 & C_3 \end{vmatrix}$$

where $\vec{A}.(\vec{B}\times\vec{C})$ is a scalar quantity.

Rack your Brain

If two vectors $\vec{A}$ and $\vec{B}$ are parallel to each other, then their Dot and Cross Product will be?

Properties of Scalar triple Product:

1. $\vec{A}.(\vec{B}\times\vec{C}) = \vec{B}.(\vec{C}\times\vec{A}) = \vec{C}.(\vec{A}\times\vec{B})$
2. $[\hat{i}\ \hat{j}\ \hat{k}] = [\hat{j}\ \hat{k}\ \hat{i}] = [\hat{k}\ \hat{i}\ \hat{j}] = 1$
3. $[\hat{i}\ \hat{k}\ \hat{j}] = -1$
4. Magnitude of scalar triple product $\vec{A}.(\vec{B}\times\vec{C})$ of three vectors is equal to the volume of a parallelopiped having sides $\vec{A}$, $\vec{B}$ and $\vec{C}$.
5. $\vec{A}.(\vec{A}\times\vec{C}) = 0$

   $\vec{C}.(\vec{A}\times\vec{C}) = 0$
6. If $\vec{A}.(\vec{B}\times\vec{C}) = 0$, then $\vec{A}$, $\vec{B}$ and $\vec{C}$ are said to be coplanar.

Vector Triple Product:

$$\vec{A}\times(\vec{B}\times\vec{C}) = \vec{B}(\vec{A}\cdot\vec{C}) - \vec{C}(\vec{A}\cdot\vec{B})$$

$(\vec{A}\times\vec{B})\times\vec{C}$ is a vector quantity.

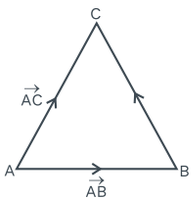
Properties of Vector Triple Product:

1. $\vec{A}\times(\vec{B}\times\vec{C}) + \vec{B}\times(\vec{C}\times\vec{A}) + \vec{C}\times(\vec{A}\times\vec{B}) = 0$
2. $\vec{A}\times(\vec{B}\times\vec{C}) \neq (\vec{A}\times\vec{B})\times\vec{C}$

Physical Significance of Cross Product:

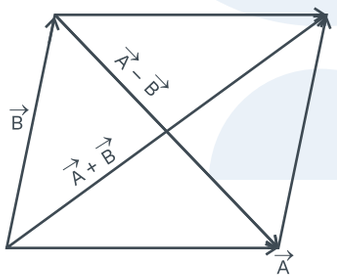
1. Area of a triangle:
   Let's consider a triangle ABC, the area of triangle is given by

$$= \frac{1}{2} | \vec{AB} \times \vec{AC} |$$



Significance of Cross Product Vector for Triangle

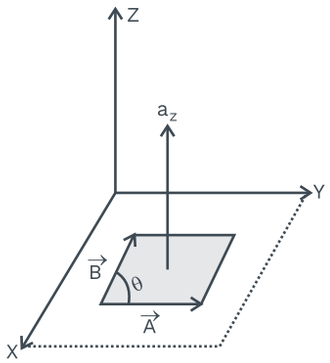
2. Area of a parallelogram
   Area = $| \vec{A} \times \vec{B} |$
   Area = $\frac{1}{2} | (\vec{A} + \vec{B}) \times (\vec{A} - \vec{B}) |$



Significance of Cross Product Vector for Triangle

Concept of Area Vector:

Consider two vectors $\vec{A}$ & $\vec{B}$



Area Vector

$\vec{A} \times \vec{B} = AB \sin\theta \hat{a}_n$ and Magnitude $|\vec{A} \times \vec{B}| = AB \sin\theta$. Now, the vectors $\vec{A}$ and $\vec{B}$ making a parallelogram, the shaded region will be become $AB \sin\theta$ and $\hat{a}_n$ will become $\hat{a}_z$. Such that, $\vec{A} \times \vec{B}$ is the magnitude which is obtained by the area of a parallelogram formed in plane in which $\vec{A}$ and $\vec{B}$ are present. The direction of $\vec{A} \times \vec{B}$ is perpendicular to the area which is following Right hand rule.

Definitions

Orthogonal:
An orthogonal system is one in which the coordinate surfaces are mutually perpendicular.

ELECTROSTATICS-I

Introduction to Electromagnetism

- Before the discovery of electromagnetism, the inventors and people used to consider electricity and magnetism as two separate topics.
- In 1873, James Clerk Maxwell published a edition called "A Treatise of Electricity and Magnetism" and changed the point of view of many people.
- His edition states that "The interactions of positive and negative charges are mediated by one force". This observation laid groundwork for Electromagnetism.
- Many scientists like Michael Faraday, Oliver Heaviside, and Heinrich Hertz contributed their ideas later to the field regarding the study of electromagnetism.
- J.C. Maxwell took the work of Faraday and other inventors and compiled it in a set of equations (popularly known as Maxwell's equations) which are recognized as the foundation of all modern theories of electromagnetic phenomena.

Gray Matter Alert!!!

Father of Electromagnetism:



Michael Faraday was the British scientist who contributed to the study of electromagnetism and electrochemistry. His main findings involve the principles underlying electromagnetic induction, electrolysis, and diamagnetism.

INTRODUCTION TO VECTOR ANALYSIS

- The distribution of a quantity in the space is represented by a field and to quantify the field, 3-D representation plays a significant role and can be done easily by vector analysis.
- Vector analysis is a mathematical tool in which electromagnetic (EM) theories are most suitably expressed and best explained.
- Scalar – A quantity that is described by the magnitude only e.g. time, mass, distance, temperature, entropy, and electric potential.
- Vector – A quantity that has both magnitude and direction e.g., velocity, force, displacement, and electric field intensity.
- A quantity in the space can be either a vector or a scalar.

COORDINATE SYSTEM

- The coordinate system sets the points of reference from which specific vector's directions may be defined.

Definitions

Electromagnetism is a branch of physics which includes the study of physical interaction (electromagnetic force) that occurs between electrically charged particles. The electro-magnetic force is conducted by the electromagnetic fields that are composed of electric field and magnetic field.

- There are spatial variations in the quantities. So, to characterize all the points, there must be some way. This demands a relevant coordinate system.
- Representation of a point or a vector can be done with any curvilinear coordinate system that can be orthogonal or non-orthogonal.
- An orthogonal coordinate system has mutually perpendicular coordinate axes.
- The three types of coordinate systems are : -
  (1) Cartesian coordinate system
  (2) Cylindrical coordinate system
  (3) Spherical coordinate system.

Definitions

The coordinate system is a system that uses "coordinates" specially for determining the position of the points or other geometric elements in Euclidean space.

CARTESIAN OR RECTANGULAR COORDINATE SYSTEM

- This system consists of x, y and z coordinate axes which are orthogonal to each other.
- A_x, A_y and A_z are the components of the vector $\vec{A}$ in cartesian coordinates x, y and z respectively.
- $\vec{A} = A_x\hat{a}_x + A_y\hat{a}_y + A_z\hat{a}_z$ where $\hat{a}_x$, $\hat{a}_y$ and $\hat{a}_z$ are the unit vectors in corresponding directions.
- **Dot product:** The dot product of two vectors $\vec{A}$ and $\vec{B}$, written as $\vec{A}\cdot\vec{B}$ is defined geometrically as the product of the magnitudes of $\vec{A}$, $\vec{B}$ and the cosine of the smaller angle between $\vec{A}$ and $\vec{B}$. Mathematically, $\vec{A}\cdot\vec{B} = |\vec{A}||\vec{B}|\cos\theta$
- If $\vec{A} = (A_x, A_y, A_z)$ and $\vec{B} = (B_x, B_y, B_z)$, then this is obtained by multiplying the component of both vectors i.e. $\vec{A}\cdot\vec{B} = A_xB_x + A_yB_y + A_zB_z$.
- Two vectors $\vec{A}$ and $\vec{B}$ are said to be orthogonal (or perpendicular) with each other if $\vec{A}\cdot\vec{B} = 0$.

Properties:

(i) Commutative law: $\vec{A}\cdot\vec{B} = \vec{B}\cdot\vec{A}$

(ii) Distributive law: $\vec{A}\cdot(\vec{B}+\vec{C}) = \vec{A}\cdot\vec{B} + \vec{A}\cdot\vec{C}$

(iii) Dot product of same vectors: $\vec{A}\cdot\vec{A} = |\vec{A}| = A^2$

(iv) $\hat{a}_x\cdot\hat{a}_y = \hat{a}_y\cdot\hat{a}_z = \hat{a}_z\cdot\hat{a}_x = 0$

$\hat{a}_x\cdot\hat{a}_x = \hat{a}_y\cdot\hat{a}_y = \hat{a}_z\cdot\hat{a}_z = 1$

Rack your Brain

How do you define cartesian equation?

- **Cross product:** The cross product of two vectors $\vec{A}$ and $\vec{B}$, written as $\vec{A} \times \vec{B}$, is a vector quantity whose magnitude is the area of the parallelogram formed by the two vectors.

  Mathematically, $\vec{A} \times \vec{B} = |\vec{A}||\vec{B}| \sin\theta \, \hat{a}_n$

  where $\hat{a}_n$ is a unit vector that is normal to the plane containing $\vec{A}$ and $\vec{B}$.
- If $\vec{A} = (A_x, A_y, A_z)$ and $\vec{B} = (B_x, B_y, B_z)$, then this is obtained by determinant or "crossing" terms in the cyclic permutation.

$$\vec{A} \times \vec{B} = \begin{vmatrix} \hat{a}_x & \hat{a}_y & \hat{a}_z \\ A_x & A_y & A_z \\ B_x & B_y & B_z \end{vmatrix} \text{ or } \vec{A} \times \vec{B} = (A_y B_z - A_z B_y)\hat{a}_x + (A_z B_x - A_x B_z)\hat{a}_y + (A_x B_y - A_y B_x)\hat{a}_z$$

- Properties:

  (i) Anti-commutative law: $\vec{A} \times \vec{B} = -(\vec{B} \times \vec{A})$

  (ii) Distributive law: $\vec{A} \times (\vec{B} + \vec{C}) = \vec{A} \times \vec{B} + \vec{A} \times \vec{C}$

  (iii) Cross-product of same vectors: $\vec{A} \times \vec{A} = 0$

  $\hat{a}_x \times \hat{a}_y = \hat{a}_z$

  (iv) $\hat{a}_y \times \hat{a}_z = \hat{a}_x$

  $\hat{a}_z \times \hat{a}_x = \hat{a}_y$
- Coordinate's axes x, y, and z have range as:
  $-\infty < x < \infty$; $-\infty < y < \infty$; $-\infty < z < \infty$

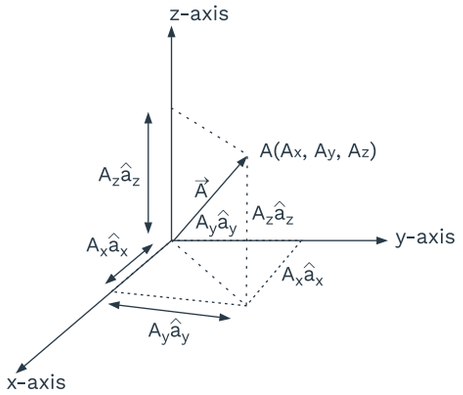


Figure 1.1: Vector $\vec{A}$ representation in Cartesian coordinate system.

Gray Matter Alert!!!

René Descartes transformed mathematics in the 17th century by presenting the first systematic link between Euclidean geometry and algebra and intiated the invention of cartesian coordinates.

CYLINDRICAL COORDINATE SYSTEM

- This is also referred to as a circular coordinate system.
- A point in cylindrical coordinate system is represented as (ρ, ϕ, z)

where,

ρ – the axial or radial distance from the z–axis to the point A or radius of cylinder passing through A.

ϕ – the azimuthal angle measured on the chosen plane (i.e., x-y plane) and the line from the origin to the projection of A on the plane.

z – the axial coordinate z is the distance from chosen plane (i.e., x-y plane) to the point A or same as z–coordinate in cartesian system.

- In cylindrical coordinate system, the component of vector $\vec{A}$ can be expressed as (A_ρ, A_ϕ, A_z)
- Vector $\vec{A}$ in cylindrical coordinate system is written as -

$$\vec{A} = A_\rho \hat{a}_\rho + A_\phi \hat{a}_\phi + A_z \hat{a}_z$$

where $\hat{a}_\rho$, $\hat{a}_\phi$ and $\hat{a}_z$ are the unit vectors in ρ, ϕ, and z directions respectively.

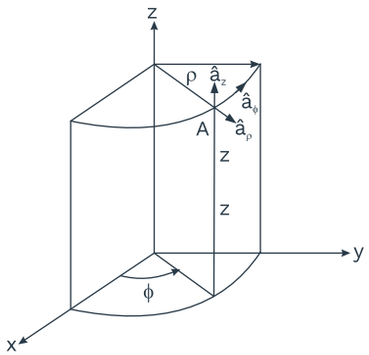


Figure 1.2: Point A and unit vectors in cylindrical coordinate system

- The cylindrical coordinate's axes lie in range of:

$0 \le \rho < \infty$

$0 \le \phi \le 2\pi$

$-\infty < z < \infty$

Gray Matter Alert!!!

Cylindrical coordinates are useful for detailed analysis of objects and phenomenon which has some rotational symmetry about the longitudinal axis, such as the electromagnetic fields produced by an electric current in a long-straight wire.

Rack your Brain

What are cylindrical harmonics or cylindrical functions and determine its relationship with the Laplacian Equation?

- The magnitude of vector $\vec{A}$ is given as-

$$\left|\vec{A}\right| = \sqrt{A_\rho^2 + A_\phi^2 + A_z^2}$$

- As the coordinate system is orthogonal in nature, thus all unit vectors will be mutually perpendicular to each other.
Hence,

$$\left.\begin{aligned} \hat{a}_\rho \cdot \hat{a}_\rho = \hat{a}_\phi \cdot \hat{a}_\phi = \hat{a}_z \cdot \hat{a}_z = 1 \\ \hat{a}_\rho \cdot \hat{a}_\phi = \hat{a}_\phi \cdot \hat{a}_z = \hat{a}_z \cdot \hat{a}_\rho = 0 \end{aligned}\right\} \text{Dot Product}$$

$$\left.\begin{aligned} \hat{a}_\rho \times \hat{a}_\phi = \hat{a}_z \\ \hat{a}_\phi \times \hat{a}_z = \hat{a}_\rho \\ \hat{a}_z \times \hat{a}_\rho = \hat{a}_\phi \end{aligned}\right\} \text{Cross Product}$$

SPHERICAL COORDINATE SYSTEM

- Spherical coordinate system is most suitable for problems with a degree of spherical symmetry.
- A point in spherical coordinate system is represented as (r, θ, ϕ).
- The vector $\vec{A}$ in spherical coordinates can be defined as $\left(A_r, A_\theta, A_\phi\right)$ or $\vec{A} = A_r \hat{a}_r + A_\theta \hat{a}_\theta + A_\phi \hat{a}_\phi$

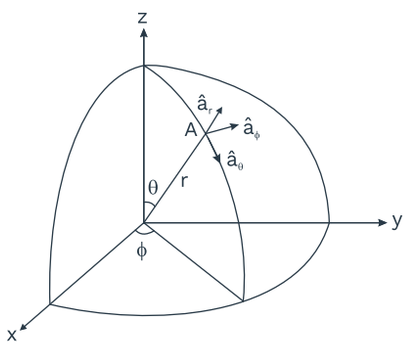


Figure 1.3: Point A and unit vectors in spherical coordinates

- Here, the coordinate axis are: -
r – is radial distance from origin to point A.
θ – is the angle between z-axis and position vector of $\vec{A}$ and called as inclination (or polar angle).
ϕ – Angle between x-axis and projection plane.

Gray Matter Alert!!!

Spherical coordinates helps to analyze systems that have some degree of symmetry about a point, such as volume integrals inside a sphere, e.g. the potential or electric field calculation in a region surrounding.

Rack your Brain

What is the Jacobian for cylindrical coordinates?

- The range of spherical coordinate axes are -
  $0 \le r < \infty$
  $0 \le \theta \le \pi$
  $0 \le \phi \le 2\pi$.
- The magnitude of $\vec{A}$ in spherical coordinate system is-

$$\left|\vec{A}\right| = \sqrt{A_r^2 + A_\theta^2 + A_\phi^2}$$

- Also, the unit vectors are mutually perpendicular in nature as this coordinate system is orthogonal.
  Therefore,

$$\left.\begin{aligned} \hat{a}_r \cdot \hat{a}_r = \hat{a}_\theta \cdot \hat{a}_\theta = \hat{a}_\phi \cdot \hat{a}_\phi = 1 \\ \hat{a}_r \cdot \hat{a}_\theta = \hat{a}_\theta \cdot \hat{a}_\phi = \hat{a}_\phi \cdot \hat{a}_r = 0 \end{aligned}\right\} \text{Dot product}$$

$$\left.\begin{aligned} \hat{a}_r \times \hat{a}_\theta = \hat{a}_\phi \\ \hat{a}_\theta \times \hat{a}_\phi = \hat{a}_r \\ \hat{a}_\phi \times \hat{a}_r = \hat{a}_\theta \end{aligned}\right\} \text{Cross Product}$$

COORDINATE SYSTEM CONVERSIONS

(1) Relationship between Cartesian and Cylindrical coordinates

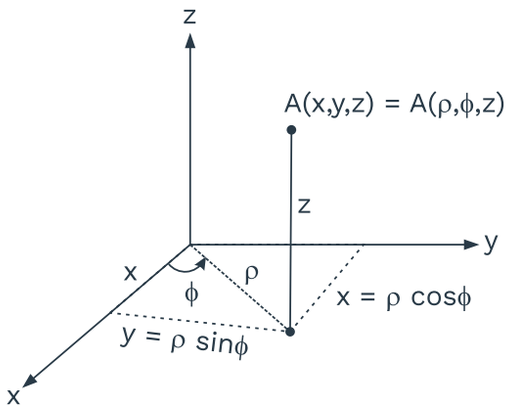


Figure 1.4: Cartesian and Cylindrical Coordinates relationship

- From the above figure 1.4, the conversion of cylindrical coordinates (ρ, ϕ, z) can be obtained as follows:

Rack your Brain

If $\vec{V} = xz\hat{i} - xy\hat{j} + yz\hat{k}$, express $\vec{V}$ in cylindrical coordinates.

Gray Matter Alert!!!

For the inter-conversion between cylindrical and cartesian coordinates, try to assume the reference plane x-y plane (with equation z = 0) and the cylindrical axis same as the cartesian z-axis.

Rack your Brain

If a point 'V' has spherical coordinates (r=5000, θ=0°, ϕ=0°), evaluate its coordinates in cartesian system?

$$x = \rho \cos\phi;\ y = \rho \sin\phi;\ z = z$$

and $\rho = \sqrt{x^2 + y^2};\ \phi = \tan^{-1}\left(\frac{y}{x}\right);\ z = z$

- Relationship between unit vectors of both coordinate systems are as follows: -

$$\hat{a}_x = \cos\phi\,\hat{a}_\rho - \sin\phi\,\hat{a}_\phi$$

$$\hat{a}_y = \sin\phi\,\hat{a}_\rho + \cos\phi\,\hat{a}_\phi$$

$$\hat{a}_z = \hat{a}_z$$

$$\hat{a}_\rho = \cos\phi\,\hat{a}_x + \sin\phi\,\hat{a}_y$$

$$\hat{a}_\phi = -\sin\phi\,\hat{a}_x + \cos\phi\,\hat{a}_y$$

$$\hat{a}_z = \hat{a}_z$$

- $\vec{A} = \left(A_x \cos\phi + A_y \sin\phi\right)\hat{a}_\rho + \left(-A_x \sin\phi + A_y \cos\phi\right)\hat{a}_\phi + A_z\hat{a}_z$

$$A_\rho = A_x \cos\phi + A_y \sin\phi$$

$$A_\phi = -A_x \sin\phi + A_y \cos\phi$$

$$A_z = A_z$$

- In matrix form, $\left(A_x, A_y, A_z\right) \rightarrow \left(A_\rho, A_\phi, A_z\right)$

$$\begin{bmatrix} A_\rho \\ A_\phi \\ A_z \end{bmatrix} = \begin{bmatrix} \cos\phi & \sin\phi & 0 \\ -\sin\phi & \cos\phi & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} A_x \\ A_y \\ A_z \end{bmatrix}$$

- Inverse conversion:

$$\left(A_\rho, A_\phi, A_z\right) \rightarrow \left(A_x, A_y, A_z\right)$$

$$\begin{bmatrix} A_x \\ A_y \\ A_z \end{bmatrix} = \begin{bmatrix} \cos\phi & -\sin\phi & 0 \\ +\sin\phi & \cos\phi & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} A_\rho \\ A_\phi \\ A_z \end{bmatrix}$$

and $\begin{bmatrix} A_x \\ A_y \\ A_z \end{bmatrix} = \begin{bmatrix} \hat{a}_x \cdot \hat{a}_\rho & \hat{a}_x \cdot \hat{a}_\phi & \hat{a}_x \cdot \hat{a}_z \\ \hat{a}_y \cdot \hat{a}_\rho & \hat{a}_y \cdot \hat{a}_\phi & \hat{a}_y \cdot \hat{a}_z \\ \hat{a}_z \cdot \hat{a}_\rho & \hat{a}_z \cdot \hat{a}_\phi & \hat{a}_z \cdot \hat{a}_z \end{bmatrix} \begin{bmatrix} A_\rho \\ A_\phi \\ A_z \end{bmatrix}$

Previous Year's Questions

$\vec{A} = \rho(z^2 - 1)\hat{a}_\rho - \rho z \cos\phi\,\hat{a}_\phi + \rho^2 z^2 \hat{a}_z$

$\vec{B} = r^2 \cos\phi\,\hat{a}_r + 2r \sin\phi\,\hat{a}_\phi$

At point D (-3, 2, 1), calculate $\vec{A}$ and $\vec{B}$ and the unit vectors in spherical coordinates perpendicular to both $\vec{A}$ and $\vec{B}$ at D.

Rack your Brain

Transform the spherical coordinates $\vec{B} = (10 / r)\hat{i} + (10\cos\theta)\hat{j} + \hat{k}$ into cylindrical coordinate system at point P(5, π/2, -2).

(2) Relationship between Cartesian and Spherical Coordinate System

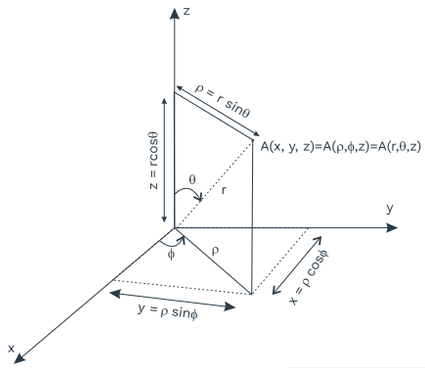


Figure 1.5: Relationship between Cartesian and Spherical Coordinate System

- The conversion of cartesian coordinate system (x, y, z) to spherical coordinates (r, θ, ϕ) is given as follows:

$$r = \sqrt{x^2 + y^2 + z^2}$$

$$\theta = \tan^{-1}\left[\frac{\sqrt{x^2 + y^2}}{z}\right]$$

$$\phi = \tan^{-1}\left(\frac{y}{x}\right)$$

$$x = r\sin\theta\cos\phi$$

$$y = r\sin\theta\sin\phi$$

$$z = r\cos\theta$$

- The Vector $\vec{A}$ can be expressed as,

$$\vec{A} = \left(A_x \sin\theta\cos\phi + A_y \sin\theta\sin\phi + A_z \cos\theta\right)\hat{a}_r + \left(A_x \cos\theta\cos\phi + A_y \cos\theta\sin\phi - A_z \sin\theta\right)\hat{a}_\theta + \left(-A_x \sin\phi + A_y \cos\phi\right)\hat{a}_\phi$$

$$A_r = A_x \sin\theta\cos\phi + A_y \sin\theta\sin\phi + A_z \cos\theta$$

$$A_\theta = A_x \cos\theta\cos\phi + A_y \cos\theta\sin\phi - A_z \sin\theta$$

$$A_\phi = -A_x \sin\phi + A_y \cos\phi$$

Gray Matter Alert!!!

The points on the coordinate plane can be varied by changing any shape or adding different colour changes. The scanner and photocopier machines also uses coordinate geometry to create the exact image provided to it.

Rack your Brain

Are all coordinates always real?

- In Matrix form:

$$(A_x, A_y, A_z) \rightarrow (A_r, A_\theta, A_\phi)$$

$$\begin{bmatrix} A_r \\ A_\theta \\ A_\phi \end{bmatrix} = \begin{bmatrix} \sin\theta\cos\phi & \sin\theta\sin\phi & \cos\theta \\ \cos\theta\cos\phi & \cos\theta\sin\phi & -\sin\theta \\ -\sin\phi & \cos\phi & 0 \end{bmatrix} \begin{bmatrix} A_x \\ A_y \\ A_z \end{bmatrix}$$

- Inverse transform:

$$(A_r, A_\theta, A_\phi) \rightarrow (A_x, A_y, A_z)$$

$$\begin{bmatrix} A_x \\ A_y \\ A_z \end{bmatrix} = \begin{bmatrix} \sin\theta\cos\phi & \cos\theta\cos\phi & -\sin\phi \\ \sin\theta\sin\phi & \cos\theta\sin\phi & \cos\phi \\ \cos\theta & -\sin\theta & 0 \end{bmatrix} \begin{bmatrix} A_r \\ A_\theta \\ A_\phi \end{bmatrix}$$

- Similarly, the unit vectors are also related to each other as follows: -

$$\hat{a}_x = \sin\theta\cos\phi\,\hat{a}_r + \cos\theta\cos\phi\,\hat{a}_\theta - \sin\phi\,\hat{a}_\phi$$

$$\hat{a}_y = \sin\theta\sin\phi\,\hat{a}_r + \cos\theta\sin\phi\,\hat{a}_\theta + \cos\phi\,\hat{a}_\phi$$

$$\hat{a}_z = \cos\theta\,\hat{a}_r - \sin\theta\,\hat{a}_\theta$$

or,

$$\hat{a}_r = \sin\theta\cos\phi\,\hat{a}_x + \sin\theta\sin\phi\,\hat{a}_y + \cos\theta\,\hat{a}_z$$

$$\hat{a}_\theta = \cos\theta\cos\phi\,\hat{a}_x + \cos\theta\sin\phi\,\hat{a}_y - \sin\theta\,\hat{a}_z$$

$$\hat{a}_\phi = -\sin\phi\,\hat{a}_x + \cos\phi\,\hat{a}_y$$

- Alternatively,

$$\begin{bmatrix} A_r \\ A_\theta \\ A_\phi \end{bmatrix} = \begin{bmatrix} \hat{a}_r\cdot\hat{a}_x & \hat{a}_r\cdot\hat{a}_y & \hat{a}_r\cdot\hat{a}_z \\ \hat{a}_\theta\cdot\hat{a}_x & \hat{a}_\theta\cdot\hat{a}_y & \hat{a}_\theta\cdot\hat{a}_z \\ \hat{a}_\phi\cdot\hat{a}_x & \hat{a}_\phi\cdot\hat{a}_y & \hat{a}_\phi\cdot\hat{a}_z \end{bmatrix} \begin{bmatrix} A_x \\ A_y \\ A_z \end{bmatrix}$$

VECTOR CALCULUS

- Vector calculus or vector analysis is associated with integration and differentiation of vector fields principally in 3-D Euclidean space.

Previous Year's Questions

Any vector in an arbitrary direction can be replaced by two or three vectors which are -

(A) perpendicular to each other and have the original vector as their resultant.

(B) parallel to each other and have the original vector as their resultant.

(C) arbitrary vectors which have original vectors as their resultant.

(D) it is impossible to resolve a vector.

Gray Matter Alert!!!

Vector calculus was explained from quaternion analysis by J. Willard Gibbs and Oliver Heaviside. But most of the mathematics and terminologies were organized by Gibbs and Edwin Bidwell Wilson in their book 'Vector analysis', in 1901.

- "Vector calculus" is sometimes used as a synonym for the more obvious subject of multivariable calculus which blends multi-integrations and partial differentiation.
- It is widely used to describe the electro-magnetic fields, gravitational fields and fluid flow in physics.

1. Differential elements in different coordinate systems

- The analysis of the integral in specific coordinate system needs knowledge of differential element of length, surface, and volume.

(i) Differential elements in Cartesian coordinate system (x, y, z)

- The differential length dl at a point is a vector from point S(x, y, z) to the point B(x + dx, y + dy, z + dz).

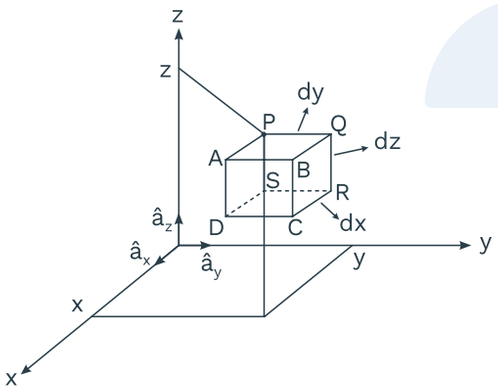


Figure 1.6: Differential elements in Cartesian coordinate system

1. Differential length is expressed as:

$$d\vec{l} = dx\,\hat{a}_x + dy\,\hat{a}_y + dz\,\hat{a}_z$$

2. Differential normal surface area is expressed as:

$$d\vec{S} = dy\ dz\,\hat{a}_x$$

$$d\vec{S} = dx\ dz\,\hat{a}_y$$

$$d\vec{S} = dx\ dy\,\hat{a}_z$$

Rack your Brain

Convert the following cartesian coordinates into cylindrical coordinates:

[sinθ, cosθ, tanθ].

Gray Matter Alert!!!

The differential element of a quantity indicates a tiny change and is described as the limit of a change in the quantity.

Previous Year's Questions

Convert the equation given in spherical coordinates into an equation in cartesian coordinates: $r^2 = 3 - \cos\phi$.

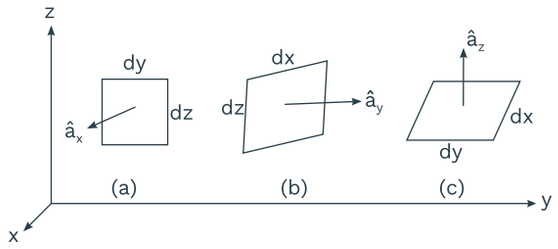


Figure 1.7: Differential Normal surface areas in Cartesian coordinate system

3. Differential volume is expressed as:

$$dV = dx\ dy\ dz$$

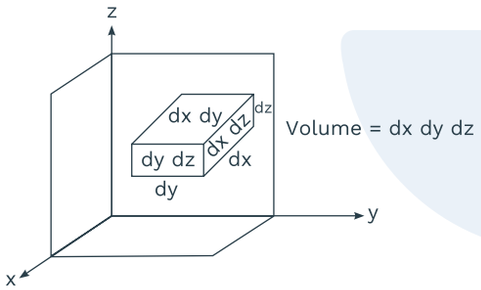


Figure 1.8: Differential volume in cartesian coordinate system

Note:

- $d\vec{l}$ and $d\vec{S}$ are vector quantities but dV is a scalar quantity.
- $d\vec{S}$ is an area of surface element and is a part that is used to describe volume also.

$$d\vec{S} = dS\ \hat{a}_n$$

Here $\hat{a}_n$ is a unit normal vector to the surface dS and directed away from the volume, if 'dS' is a part of an enclosed volume.

- For easy calculation, determining $d\vec{l}$ can lead to evaluation of $d\vec{S}$ and dV easily.
For example, $d\vec{S}$ along x-direction or $\hat{a}_x$ will be product of $d\vec{l}$ components in other two directions i.e., y and z directions.
Mathematically expressed as,

$$d\vec{S} = dy\ dz\ \hat{a}_x$$

Rack your Brain

The differential length vector is a directed distance. Thus, the units of its magnitude must be distance (e.g., meters, feet). The differential value $d\phi$ has units of radians,
Try to find the unit of the differential value $\rho d\phi$?

Previous Year's Questions

On either side of a charge-free interface between two media -

(A) the normal components of the electric field are equal.
(B) the tangential component of the electric field is equal.
(C) the normal components of the electric flux density are equal.
(D) the tangential components of the electric flux density are equal.

Similarly, $d\vec{S}$ along y and z-direction would be -

$$d\vec{S} = dx\ dz\ \hat{a}_y \text{ and } d\vec{S} = dx\ dy\ \hat{a}_z$$

(ii) Differential elements in cylindrical coordinate system (ρ, ϕ, z)

1. Differential length is expressed as -

$$d\vec{l} = d\rho\hat{a}_\rho + \rho d\phi\hat{a}_\phi + dz\hat{a}_z$$

2. Differential normal area is expressed as -

$$d\vec{S} = \rho d\phi dz\hat{a}_\rho$$
$$d\vec{S} = d\rho dz\hat{a}_\phi$$
$$d\vec{S} = \rho d\rho d\phi\hat{a}_z$$

3. Differential volume is expressed as -

$$dV = \rho\ d\rho\ d\phi\ dz$$

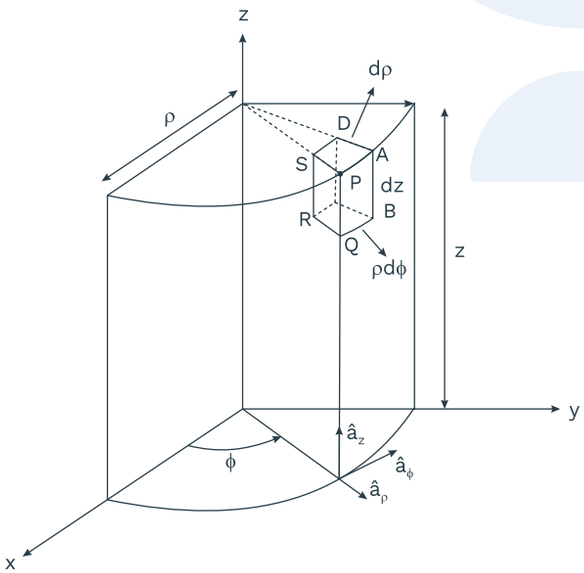


Figure 1.9: Differential elements in Cylindrical coordinate system

Rack your Brain

Given point P(–1, 3, 3) and vector $\vec{A} = y\hat{a}_x + (x+z)\hat{a}_y$. Evaluate vector $\vec{A}$ at point P in the cartesian, cylindrical, and spherical coordinate.

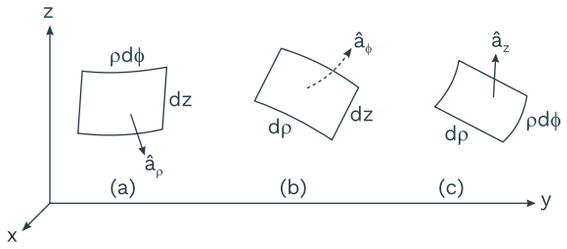


Figure 1.10: Differential normal surface in cylindrical coordinate system

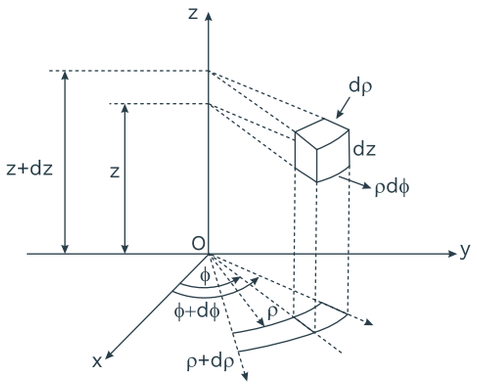


Figure 1.11: Differential volume in cylindrical coordinate system

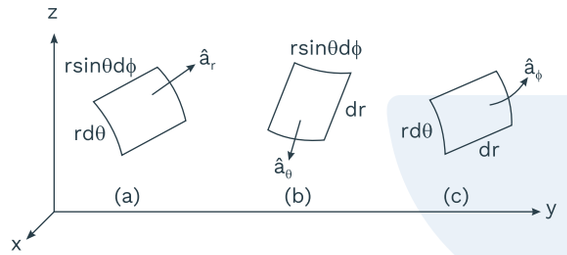


Figure 1.12: Differential surface in spherical coordinate system

(iii) Differential elements in spherical coordinate system (r, θ, ϕ)

1. Differential length is expressed as-

$$d\vec{l} = dr\,\hat{a}_r + rd\theta\,\hat{a}_\theta + r\sin\theta\,d\phi\,\hat{a}_\phi$$

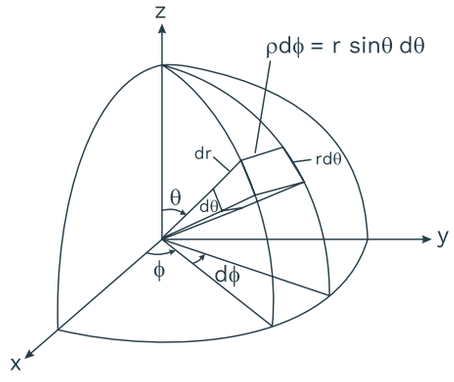


Figure 1.13a: Differential length in spherical coordinate system

Previous Year's Questions

A metal sphere with 1 m radius and a surface charge density of 10 coulombs/m^2 is enclosed in a cube of 10 m side. The total outward electric length normal to the surface of the cube is -

(A) 40 π Coulombs
(B) 10 π Coulombs
(C) 5 π Coulombs
(D) None of the above

2. Differential normal surface is expressed as:

$$d\vec{S} = r^2 \sin\theta \, d\theta \, d\phi \, \hat{a}_r$$
$$d\vec{S} = r \sin\theta \, dr \, d\phi \, \hat{a}_\theta$$
$$d\vec{S} = r \, dr \, d\theta \, \hat{a}_\phi$$

3. Differential volume is expressed as -

$$dV = r^2 \sin\theta \, dr \, d\theta \, d\phi$$

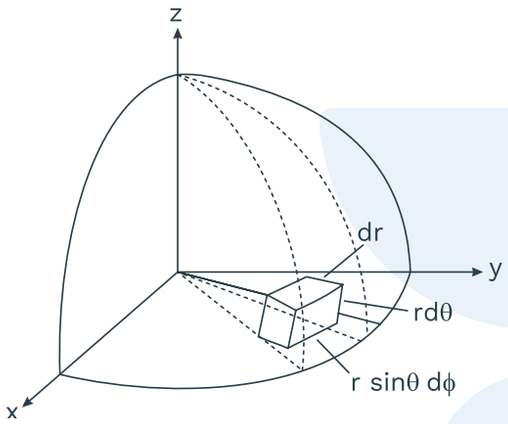


Figure 1.13b: Differential volume in spherical coordinate system

Rack your Brain

The integral of a vector is a scalar or a vector?

Gray Matter Alert!!!

Some of the vector calculus fundamental theorems involves line integrals of vector fields like Green's Theorem and Stokes' theorem which relates line integral around closed curves to double integrals (or surface integrals).

Differential Element	Types of Coordinate system		
	Cartesian (x, y, z)	Cylindrical (ρ, ϕ, z)	Spherical (r, θ, ϕ)
Length (dl)	$dx\,\hat{a}_x$ $dy\,\hat{a}_y$ $dz\,\hat{a}_z$	$d\rho\,\hat{a}_\rho$ $\rho\,d\phi\,\hat{a}_\phi$ $dz\,\hat{a}_z$	$dr\,\hat{a}_r$ $r\,d\theta\,\hat{a}_\theta$ $r\sin\theta\,d\phi\,\hat{a}_\phi$
Surface (dS)	$dy\,dz\,\hat{a}_x$ $dx\,dz\,\hat{a}_y$ $dx\,dy\,\hat{a}_z$	$\rho\,d\phi\,dz\,\hat{a}_\rho$ $d\rho\,dz\,\hat{a}_\phi$ $\rho\,d\rho\,d\phi\,\hat{a}_z$	$r^2\sin\theta\,d\theta\,d\phi\hat{a}_r$ $r\sin\theta\,dr d\phi\,\hat{a}_\theta$ $r\,dr\,d\theta\,\hat{a}_\phi$
Volume (dV)	$dx\,dy\,dz$	$r d\rho\,d\phi\,dz$	$r^2\sin\theta\,dr\,d\theta\,d\phi$

Table 1.1 : Representation of differential elements in all coordinate systems

VECTOR CALCULUS ELEMENTS

Line, surface, and volume integrals

- The study of charge distribution requires various types of integrals in electromagnetic theory and a charge can exist in a point form, line form, surface form or volume form.
- Types of Integrals:
  1. Line Integral
  2. Surface Integral
  3. Volume Integral

(1) Line Integral

- A line can denote a straight line, or it can be a path travelled along a curve.
  Hence, from the mathematical point of view, a line is a curved path in space.
- Line, contour, and curve terms are used interchangeably.
- Assume a vector field A around a curve L, then the line integral can be expressed as

$$\int_L \vec{A} \cdot d\vec{l} = \int_a^b |A| \cos\theta \, dl$$

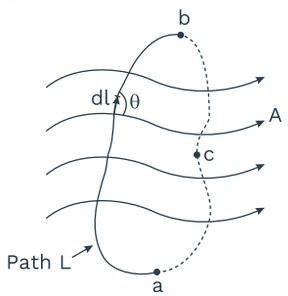


Figure 1.14: Path of integration of vector field $\vec{A}$ around path L

- If the path of integration is a closed curve abca [as mentioned in figure (1.14)], then the integral becomes a closed contour integral or is called circulation of A around L.
- Mathematically, closed contour integral is expressed as-

$$\oint_L \vec{A} \cdot d\vec{l}$$

Previous Year's Questions

Suppose an object moves from (-1, 1) to (2, 4) along the path $r(t) = (t, t^2)$, subject to the force $\vec{F} = (x \sin y, y)$. Find the work done.

- Also, if a charge exists along a line [as shown in figure (1.15)], then the total charge is calculated by obtaining the line integral.

$$\therefore\ Q = \int_L \rho_L dl$$

where, ρ_L = Line charge density [or charge per unit length (C/m)], Q = Charge (C).

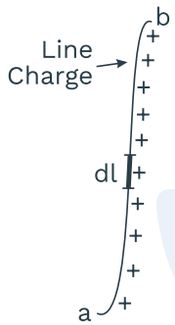


Figure 1.15: Charge distributed along a line

(2) Surface Integral

- A charge may be spread over a surface [as shown in figure (1.16)] and exists in a distributed form. Similarly, a flux may pass through the surface.
- Analyzing such instances, an integral called surface integral is required to be carried out over a surface related to the vector field.
- For a charge distributed over a smooth surface S(a continuous region), the total charge distributed on surface S will be defined using surface integral.

$$\text{i.e. } Q = \int_S \rho_S dS$$

where, ρ_S = surface charge density (C/m^2)
dS = elementary surface

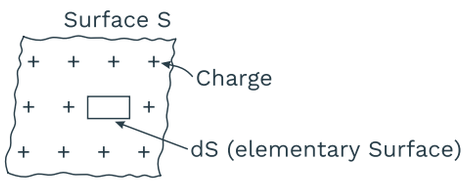


Figure 1.16: Charge distributed along a surface

Previous Year's Questions

Find the gradient vector field of the following function.

$f(x, y) = x^2 \sin(5y)$

Gray Matter Alert!!!

A line integral is specified for functions of two or more variables, and the interval of integration is replaced by a curve connecting the two endpoints of the interval. In a surface integral, the curve is replaced by a surface in three-dimensional space.

- And the total flux $\vec{A}$ crossing the surface S can be expressed as

$$\psi = \int_S |\vec{A}| \cos\theta \, d\vec{S} = \int_S \vec{A} \cdot \hat{a}_n \, dS$$

or $\psi = \int_s \vec{A} \cdot d\vec{S}$

where, $\hat{a}_n$ = unit vector normal to surface S.

- For the closed surface (can be assumed as volume), the above equation becomes -

$$\psi = \oint_s \vec{A} \cdot d\vec{S}$$

where, ψ = the net outward flux of $\vec{A}$ from surface S.

Note:

(1) Open surface is referred as a closed path surface, but the closed surface is referred as volume.

(2) Mathematically, the surface integral involves the double integration procedure.

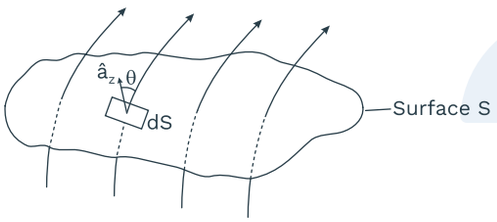


Figure 1.17: Flux of a vector field a through surface

(3) Volume Integral

- If the charge distribution exists in 3-Dimensional volume, then to calculate the total charge, the volume integral is required.
- The volume integral of the scalar ρ_V over the volume V is expressed as

$$Q = \int_V \rho_V \, dV$$

where, ρ_V = volume charge density
dV = elementary volume

Gray Matter Alert!!!

If at a point P, the gradient of a function is non-zero, then the gradient's direction will be in the direction where rate of change of function is maximum from P. The magnitude of the gradient will be the rate of increase in that direction.

Rack your Brain

Curl of gradient of a vector would be always_____.

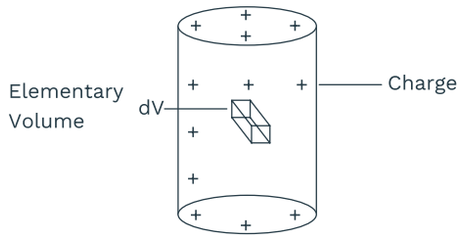


Figure 1.18: Charge distributed over the volume

Rack your Brain

Does del operator has a unit?

Del Operator (∇)

- The del operator is the vector differential operator and it is denoted by symbol '∇' generally pronounced as del.
- In Cartesian coordinates,

$$\nabla = \frac{\partial}{\partial x}\hat{a}_x + \frac{\partial}{\partial y}\hat{a}_y + \frac{\partial}{\partial z}\hat{a}_z$$

- This vector differential operator is referred to as gradient operator.
- The del operator has different meaning depending upon its use:

(1) When del operator (∇) is applied to a scalar function, we get gradient of scalar V and is written as ∇V.

(2) The dot product of del operator (∇) and vector $\vec{A}$, is regarded as the divergence of vector $\vec{A}$ and is written as: $\nabla \cdot \vec{A}$

(3) The cross product of del operator (∇) and vector $\vec{A}$ is referred as the curl of vector $\vec{A}$ and is written as: $\nabla \times \vec{A}$

(4) The dot product of del operator and gradient of scalar V is regarded as the Laplacian of scalar field V and is written as: $\nabla \cdot (\nabla V) = \nabla^2 V$

Gray Matter Alert!!!

The gradient is the null vector at a stationary point (where the derivative vanishes). So the gradient is applied in optimization theory to maximize a function.

Del operator in different coordinate systems

- Cartesian coordinates system (x, y, z):

$$\nabla = \frac{\partial}{\partial x}\hat{a}_x + \frac{\partial}{\partial y}\hat{a}_y + \frac{\partial}{\partial z}\hat{a}_z$$

- Cylindrical coordinates system (ρ, ϕ, z):

$$\nabla = \frac{\partial}{\partial \rho}\hat{a}_\rho + \frac{1}{\rho}\frac{\partial}{\partial \phi}\hat{a}_\phi + \frac{\partial}{\partial z}\hat{a}_z$$

Rack your Brain

Where do we use gradient in practical life?

- Spherical coordinates system (r, θ, ϕ):

$$\nabla = \frac{\partial}{\partial r}\hat{a}_r + \frac{1}{r}\frac{\partial}{\partial \theta}\hat{a}_\theta + \frac{1}{r\sin\theta}\frac{\partial}{\partial \phi}\hat{a}_\phi$$

Gradient of a scalar (∇V)

- The gradient of a scalar field V is a vector function that represents both magnitude and direction.
- Depending upon the position where gradient is evaluated, it may have different magnitudes & directions located in the space.
- Mathematical expressions of gradient of scalar V in different coordinate systems are described below -
- Cartesian coordinate system (x, y, z):

$$\nabla V = \left(\frac{\partial}{\partial x}\hat{a}_x + \frac{\partial}{\partial y}\hat{a}_y + \frac{\partial}{\partial z}\hat{a}_z\right)V$$

or $$\nabla V = \frac{\partial V}{\partial x}\hat{a}_x + \frac{\partial V}{\partial y}\hat{a}_y + \frac{\partial V}{\partial z}\hat{a}_z$$

- Cylindrical coordinate system (ρ, ϕ, z):

$$\nabla V = \left(\frac{\partial}{\partial \rho}\hat{a}_\rho + \frac{1}{\rho}\frac{\partial}{\partial \phi}\hat{a}_\phi + \frac{\partial}{\partial z}\hat{a}_z\right)V$$

or $$\nabla V = \frac{\partial V}{\partial \rho}\hat{a}_\rho + \frac{1}{\rho}\frac{\partial V}{\partial \phi}\hat{a}_\phi + \frac{\partial V}{\partial z}\hat{a}_z$$

- Spherical coordinate system (r, θ, ϕ):

$$\nabla V = \left(\frac{\partial}{\partial r}\hat{a}_r + \frac{1}{r}\frac{\partial}{\partial \theta}\hat{a}_\theta + \frac{1}{r\sin\theta}\frac{\partial}{\partial \phi}\hat{a}_\phi\right)V$$

or $$\nabla V = \frac{\partial V}{\partial r}\hat{a}_r + \frac{1}{r}\frac{\partial V}{\partial \theta}\hat{a}_\theta + \frac{1}{r\sin\theta}\frac{\partial V}{\partial \phi}\hat{a}_\phi$$

- Following identities/formulas on gradient are widely useful for simplification of problems:

(i) $\nabla (V + U) = \nabla V + \nabla U$

(ii) $\nabla (VU) = V\nabla U + U\nabla V$

(iii) $\nabla\left[\frac{V}{U}\right] = \frac{U\nabla V - V\nabla U}{U^2}$

(iv) $\nabla V^n = nV^{n-1}\nabla V$

where U and V are scalar quantities and n is an integer.

Rack your Brain

If divergence and curl are zero, what will be the nature of the field?

Gray Matter Alert!!!

Divergence will be zero for a constant field (where the number of lines entering and exiting from a volume is the same) for all objects placed entirely in the field.

Example 7:
Determine the gradient of the following scalar fields:

(i) $V = 20e^{-x}\sin\left(\frac{\pi y}{6}\right)$ at P (0, 1, 1)

(ii) $W = 25\,\rho\sin\phi$ at $Q\left(\sqrt{2}, \frac{\pi}{2}, 5\right)$

(iii) $T = \frac{40\cos\theta}{r^2}$ at R(3, 60°, 30°)

Gray Matter Alert!!!

A vector field with curl zero is called as irrotational field.

Solution:
(i) For Cartesian coordinates the gradient will be

$$\nabla V = \frac{\partial V}{\partial x}\hat{a}_x + \frac{\partial V}{\partial y}\hat{a}_y + \frac{\partial V}{\partial z}\hat{a}_z$$

$$\& \quad \frac{\partial V}{\partial x} = \frac{\partial}{\partial x}\left[20e^{-x}\sin\left(\frac{\pi y}{6}\right)\right] = -20e^{-x}\sin\left(\frac{\pi y}{6}\right)$$

$$\frac{\partial V}{\partial y} = \frac{\partial}{\partial y}\left[20e^{-x}\sin\left(\frac{\pi y}{6}\right)\right] = 20e^{-x}\cos\left(\frac{\pi y}{6}\right)\cdot\frac{\pi}{6}$$

$$\frac{\partial V}{\partial z} = \frac{\partial}{\partial z}\left[20e^{-x}\sin\left(\frac{\pi y}{6}\right)\right] = 0$$

$$\nabla V = -20e^{-x}\sin\left(\frac{\pi y}{6}\right)\hat{a}_x + \frac{20\pi e^{-x}}{6}\cos\left(\frac{\pi y}{6}\right)\hat{a}_y + 0$$

At P (0, 1, 1)

$$\nabla V|_{(0,1,1)} = -10\hat{a}_x + 9.06\hat{a}_y$$

(ii) $W = 25\,\rho\sin\phi$ in cylindrical coordinates system.

$$\nabla W = \frac{\partial W}{\partial \rho}\hat{a}_\rho + \frac{1}{\rho}\frac{\partial W}{\partial \phi}\hat{a}_\phi + \frac{\partial W}{\partial z}\hat{a}_z$$

$$\text{here } \frac{\partial W}{\partial \rho} = 25\sin\phi$$

$$\frac{\partial W}{\partial \phi} = 25\,\rho\cos\phi$$

$$\frac{\partial W}{\partial z} = 0$$

$$\therefore \nabla W = 25\sin\phi\,\hat{a}_\rho + 25\cos\phi\,\hat{a}_\phi$$

$$\text{at } Q\left(\sqrt{2}, \frac{\pi}{2}, 5\right)$$

$$\nabla W\Big|_{\text{at}\left(\sqrt{2},\frac{\pi}{2},5\right)} = 25\,\hat{a}_\rho$$

(iii) $T = \dfrac{40\cos\theta}{r^2}$ in spherical coordinates,

$$\nabla T = \frac{\partial T}{\partial r}\hat{a}_r + \frac{1}{r}\frac{\partial T}{\partial \theta}\hat{a}_\theta + \frac{1}{r\sin\theta}\frac{\partial T}{\partial \phi}\hat{a}_\phi$$

$$\frac{\partial T}{\partial r} = 40\cos\theta\left[-2r^{-3}\right] = -80\frac{\cos\theta}{r^3}$$

$$\frac{\partial T}{\partial \theta} = \frac{-40\sin\theta}{r^2}$$

$$\frac{\partial T}{\partial \phi} = 0$$

$$\text{and } \nabla T = \frac{-80\cos\theta}{r^3}\hat{a}_r - \frac{40}{r^2}\sin\theta\,\hat{a}_\theta$$

At given point R(3, 60°, 30°) ∇T is -

$$\nabla T\Big|_{\text{at}(3,60°,30°)} = -1.49\,\hat{a}_r - 0.93\,\hat{a}_\theta$$

Rack your Brain

Results of divergence will be always scalar or vector?

Divergence of a vector

- The divergence of vector $\vec{A}$ at point is the net outward flux from a closed surface S per unit volume (as volume shrinks at P).
- Mathematically,

$$\text{div}\,\vec{A} = \nabla\cdot\vec{A} = \lim_{\Delta V\to 0}\frac{\oint \vec{A}\cdot d\vec{S}}{\Delta V}$$

where, ΔV = volume enclosed by the closed surface in which P is located.

Definitions

The divergence theorem tells that when we combine up all the minute parts of outward flow in a volume using a triple integral of divergence, it provides us the total outward flux flows from the volume and measured through its surface.

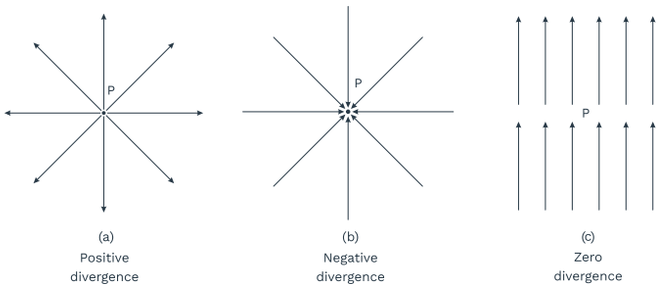


Figure 1.19: Demonstration of divergence of a vector field at P point

Gray Matter Alert!!!

The solenoidal vector field has divergence equal to zero throughout.

COORDINATE SYSTEM	DIVERGENCE OF VECTOR $\vec{A}$
Cartesian coordinates (x, y, z)	$\nabla \cdot \vec{A} = \frac{\partial A_x}{\partial x} + \frac{\partial A_y}{\partial y} + \frac{\partial A_z}{\partial z}$
Cylindrical coordinates (ρ, ϕ, z)	$\nabla \cdot \vec{A} = \frac{1}{\rho}\frac{\partial}{\partial \rho}(\rho A_\rho) + \frac{1}{\rho}\frac{\partial A_\phi}{\partial \phi} + \frac{\partial A_z}{\partial z}$
Spherical coordinates (r, θ, ϕ)	$\nabla \cdot \vec{A} = \frac{1}{r^2}\frac{\partial}{\partial r}(r^2 A_r) + \frac{1}{r \sin\theta}\frac{\partial}{\partial \theta}(A_\theta \sin\theta) + \frac{1}{r \sin\theta}\frac{\partial A_\phi}{\partial \phi}$

Table 1.2: Divergence of vector $\vec{A}$ at point P in different coordinate systems

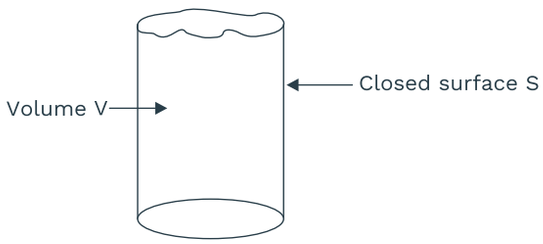


Figure 1.20: Volume V enclosed by closed surface S

Divergence Theorem

- The divergence theorem states that "the total outward flux of a vector field $\vec{A}$ through the closed surface S is similar to the volume integral of the divergence of vector $\vec{A}$".
- It is also known as Gauss-ostrogradsky theorem but commonly referred as divergence theorem.

- According to the statement above, mathematical representation of divergence theorem would be as follows:

$$\underbrace{\oint_s \vec{A}\cdot d\vec{S}}_{\text{Total outward flux of } \vec{A} \text{ through closed surface S}} = \underbrace{\int_v (\nabla\cdot\vec{A})\,dV}_{\text{Volume integral of divergence of vector } \vec{A}}$$

- The important properties of divergence of a vector field:

(i) $\nabla\cdot\left(\vec{A}+\vec{B}\right) = \nabla\cdot\vec{A} + \nabla\cdot\vec{B}$

(ii) $\nabla\cdot(V\vec{A}) = V\nabla\cdot\vec{A} + \vec{A}\cdot\nabla V$

here V is scalar and $\vec{A}$ & $\vec{B}$ are vectors.

(iii) It produces a scalar field as the scalar product is involved. (Dot product is scalar product but cross product is a vector product.)

Rack your Brain

Try to calculate the curl of a scalar field?

Example 8:

Determine the divergence of the following vector fields:

(i) $\vec{S} = x^3y^2z\hat{a}_x + x^2z\hat{a}_z$

(ii) $\vec{T} = \rho\sin\phi\,\hat{a}_\rho + \rho^2 z a_\phi + 2z\cos\phi\hat{a}_z$

(iii) $\vec{W} = \frac{1}{r^2}\cos\theta\hat{a}_r + r\sin\theta\cos\phi\,a_\theta + \cos\theta\,a_\phi$

Rack your Brain

Which of the following is not vector functions in electromagnetics?
Gradient, divergence, curl.

Solution:

According to the divergence of vectors

(i) $\nabla\cdot\vec{S} = \frac{\partial S_x}{\partial x} + \frac{\partial S_y}{\partial y} + \frac{\partial S_z}{\partial z}$ for cartesian coordinates system

$$= \frac{\partial\left(x^3y^2z\right)}{\partial x} + \frac{\partial}{\partial y}(0) + \frac{\partial}{\partial z}\left(x^2z\right)$$

$$\nabla\cdot\vec{S} \Rightarrow 3x^2y^2z + x^2$$

(ii) For cylindrical coordinate system.

$$\nabla\cdot\vec{T} = \frac{1}{\rho}\frac{\partial}{\partial\rho}\left(\rho T_\rho\right) + \frac{1}{\rho}\frac{\partial}{\partial\phi}\left(T_\phi\right) + \frac{\partial}{\partial z}\left(T_z\right)$$

$$= \frac{1}{\rho}\frac{\partial}{\partial \rho}(\rho^2 \sin\phi) + \frac{1}{\rho}\frac{\partial}{\partial \phi}(\rho^2 z) + \frac{\partial}{\partial z}(2z\cos\phi)$$

$\Rightarrow 2 \sin \phi + 0 + 2 \cos \phi$

$\Rightarrow 2 (\sin \phi + \cos \phi)$

(iii) For spherical coordinates.

$$\nabla \cdot \vec{W} = \frac{1}{r^2}\frac{\partial}{\partial r}\left(r^2 W_r\right) + \frac{1}{r\sin\theta}\frac{\partial}{\partial \theta}\left(W_\theta \sin\theta\right) + \frac{1}{r\sin\theta}\frac{\partial W_\phi}{\partial \phi}$$

$$= \frac{1}{r^2}\frac{\partial}{\partial r}(\cos\theta) + \frac{1}{r\sin\theta}\frac{\partial}{\partial \theta}(r\sin^2\theta\cos\phi)$$

$$+ \frac{1}{r\sin\theta}\frac{\partial}{\partial \phi}(\cos\theta)$$

$$= 0 + \frac{1}{r\sin\theta} \times 2r\sin\theta\cos\theta\cos\phi + 0$$

$= 2 \cos\theta \cos\phi$

Curl of a vector

- The curl of a vector can be obtained by the circulation of a vector filed around a closed path.
  Mathematically, it is expressed as-

$$\text{Curl } \vec{A} = \left(\lim_{\Delta S \to 0} \frac{\oint \vec{A} \cdot d\vec{l}}{\Delta S}\right)_{max} \hat{a}_n$$

  where ΔS = Area enclosed by the line integral in the normal direction.
  $\hat{a}_n$ = Unit vector normal to surface ΔS.
- Also, the maximum circulation of $\vec{A}$ per unit area tends to zero that has direction normal to the surface and is called as curl of the vector $\vec{A}$.
- In other words, the cross product of del operator and the vector field gives the curl of that vector.
  Symbolically, it can be represented as-

$$\nabla \times \vec{A} = \text{Curl } \vec{A}$$

Rack your Brain

If a vector field describes the linear velocities of each part of rotating disk, then the curl must be _____ at all points.

Previous Year's Questions

Given

$$\vec{V} = x\cos^2 y\hat{i} + x^2 e^z \hat{j} + z\sin^2 y\hat{k}$$

and S the surface of unit cube with one corner at the origin and edges parallel to the coordinate axis, the value of the integral $\iint_C \vec{V} \cdot \hat{n} d\vec{S}$ is _____.

- The definition of $\nabla \times \vec{A}$ is independent of the coordinates system.
- Therefore, in cartesian coordinates, the curl of $\vec{A}$ is given by-

$$\nabla \times \vec{A} = \begin{vmatrix} \hat{a}_x & \hat{a}_y & \hat{a}_z \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ A_x & A_y & A_z \end{vmatrix}$$

$$\text{Or, } \nabla \times \vec{A} = \left[\frac{\partial A_z}{\partial y} - \frac{\partial A_y}{\partial z}\right]\hat{a}_x + \left[\frac{\partial A_x}{\partial z} - \frac{\partial A_z}{\partial x}\right]\hat{a}_y$$

$$+\left[\frac{\partial A_y}{\partial x} - \frac{\partial A_x}{\partial y}\right]\hat{a}_z$$

- In cylindrical coordinates system, the curl of $\vec{A}$ is defined as-

$$\nabla \times \vec{A} = \frac{1}{\rho}\begin{vmatrix} \hat{a}_\rho & \rho\hat{a}_\phi & \hat{a}_z \\ \frac{\partial}{\partial \rho} & \frac{\partial}{\partial \phi} & \frac{\partial}{\partial z} \\ A_\rho & \rho A_\phi & A_z \end{vmatrix}$$

- $\nabla \times \vec{A} = \left[\frac{1}{\rho}\frac{\partial A_z}{\partial \phi} - \frac{\partial A_\phi}{\partial z}\right]\hat{a}_\rho + \left[\frac{\partial A_\rho}{\partial z} - \frac{\partial A_z}{\partial \rho}\right]\hat{a}_\phi$

$$+\frac{1}{\rho}\left[\frac{\partial\left(\rho A_\phi\right)}{\partial \rho} - \frac{\partial A_\rho}{\partial \phi}\right]\hat{a}_z$$

- In spherical coordinates, the curl of $\vec{A}$ is given by-

$$\nabla \times \vec{A} = \frac{1}{r^2 \sin\theta}\begin{vmatrix} \hat{a}_r & r\hat{a}_\theta & r\sin\theta\hat{a}_\phi \\ \frac{\partial}{\partial r} & \frac{\partial}{\partial \theta} & \frac{\partial}{\partial \phi} \\ A_r & rA_\theta & r\sin\theta A_\phi \end{vmatrix}$$

$$\nabla \times \vec{A} = \frac{1}{r\sin\theta}\left[\frac{\partial\left(\sin\theta A_\phi\right)}{\partial \theta} - \frac{\partial(A_\theta)}{\partial \phi}\right]\hat{a}_r$$

$$+\frac{1}{r}\left[\frac{1}{\sin\theta}\frac{\partial A_r}{\partial \phi} - \frac{\partial(rA_\phi)}{\partial r}\right]\hat{a}_\theta + \frac{1}{r}\left[\frac{\partial(rA_\theta)}{\partial r} - \frac{\partial A_r}{\partial \theta}\right]\hat{a}_\phi$$

Previous Year's Questions

Apply the divergence theorem to evaluate $\int_S \vec{A} \cdot d\vec{S}$, where $\vec{A} = x^2\hat{a}_x + y^2\hat{a}_y + z^2\hat{a}_z$ and S is the surface of the solid bounded by the cylinder ρ = 1 and planes z = 1 and z = 2

Gray Matter Alert!!!

The curl gives a sense of the rotational nature of a vector field.

Important properties of the curl:

(1) The curl of a vector field gives another vector field.

(2) The divergence of curl of a vector field is zero. Mathematically, $\nabla \cdot \left(\nabla \times \vec{A}\right) = 0$ where $\vec{A}$ is a vector field.

(3) The curl of a gradient of a scalar field is also zero, i.e., $\nabla \times (\nabla V) = 0$ where V is a scalar field.

(4) $\nabla \times \left(\vec{A} + \vec{B}\right) = \nabla \times \vec{A} + \nabla \times \vec{B}$

(5) $\nabla \times \left(\vec{A} \times \vec{B}\right) = A\left(\nabla \cdot \vec{B}\right) - \vec{B}\left(\nabla \cdot \vec{A}\right) + \left(\vec{B} \cdot \nabla\right)\vec{A} - \left(\vec{A} \cdot \nabla\right)\vec{B}$

(6) $\nabla \times \left(V\vec{A}\right) = V\nabla \times \vec{A} + \nabla V \times \vec{A}$

- As discussed above that, the curl provides the maximum value of the circulation of the field per unit area. So, the curl of a vector field $\vec{A}$ at a point Q may be referred as the measure of circulation or the amount of field curls around point Q.

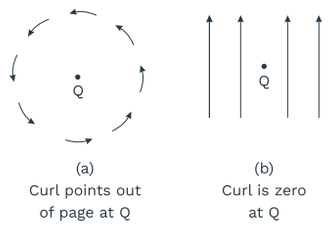


Figure 1.21: Illustration of curl

Stokes' Theorem

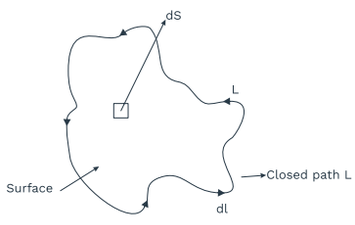


Figure 1.22: Illustration of dl and dS involved in Stokes' theorem

Rack your Brain

Are there any particular conditions required for Stokes's theorem?

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Stokes' theorem generalizes Green's theorem to 3-D from rotation in a planar region to the circulation along a surface. As Green's theorem applies only in 2-D vector fields and planes.

Stokes' Theorem

The line integral of a vector field (say $\vec{A}$) around a closed path L is equal to the surface integral of the curl of $\vec{A}$ over the open surface S bounded by and $\nabla \times \vec{A}$L, considering $\vec{A}$ are continuous on surface S.

- Mathematically the Stokes' theorem relates a closed line integral to a surface integral of a vector field $\vec{A}$ and is given by-

$$\oint_L \vec{A} \cdot d\vec{l} = \int_s (\nabla \times \vec{A}) \cdot d\vec{S}$$

Note:

Stokes' theorem is often used for inter-conversions between line integral and surface integral of a vector field whereas the divergence theorem is used for inter-conversions between surface and volume integral of a vector field.

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In the physics of electro-magnetism, Stokes' theorem justifies the equality of the differential form of the Maxwell–Faraday equation, the Maxwell–Ampere equation and the integral form of these equations.

Example 9:

Evaluate the curl of vector fields:

(i) $\vec{A} = x^2yz\hat{a}_x + xz\hat{a}_z$

(ii) $\vec{B} = \rho \sin\phi \hat{a}_\rho + \rho^2 z\hat{a}_\phi + z\cos\phi \hat{a}_z$

(iii) $\vec{C} = \frac{1}{r^2}\cos\theta \hat{a}_r + r\sin\theta\cos\phi \hat{a}_\theta + \cos\theta \hat{a}_\phi$

Solution:

(i) In cartesian coordinates,

$$\nabla \times \vec{A} = \left(\frac{\partial A_z}{\partial y} - \frac{\partial A_y}{\partial z}\right)\hat{a}_x + \left(\frac{\partial A_x}{\partial z} - \frac{\partial A_z}{\partial x}\right)\hat{a}_y + \left(\frac{\partial A_y}{\partial x} - \frac{\partial A_x}{\partial y}\right)\hat{a}_z$$

$$\nabla \times \vec{A} = (0-0)\hat{a}_x + (x^2y - z)\hat{a}_y + (0 - x^2z)\hat{a}_z$$

(ii) According to cylindrical coordinates

$$\nabla \times \vec{B} = \left[\frac{1}{\rho}\frac{\partial B_z}{\partial \phi} - \frac{\partial B_\phi}{\partial z}\right]\hat{a}_\rho + \left[\frac{\partial B_\rho}{\partial z} - \frac{\partial B_z}{\partial \rho}\right]\hat{a}_\phi$$

$$+\frac{1}{\rho}\left[\frac{\partial}{\partial\rho}\left(\rho B_\phi\right)-\frac{\partial B_\rho}{\partial\phi}\right]\hat{a}_z$$

$$=\left(\frac{-z}{\rho}\sin\phi-\rho^2\right)\hat{a}_\rho+(0-0)\,\hat{a}_\phi+\frac{1}{\rho}\left(3\rho^2 z-\rho\cos\phi\right)\hat{a}_z$$

$$=\frac{-1}{\rho}\left(z\sin\phi+\rho^3\right)\hat{a}_\rho+\left(3\rho z-\cos\phi\right)\hat{a}_z$$

(iii) According to spherical coordinates,

$$\nabla\times\vec{C}=\frac{1}{r\sin\theta}\left[\frac{\partial}{\partial\theta}(C_\phi\sin\theta)-\frac{\partial C_\theta}{\partial\phi}\right]\hat{a}_r$$

$$+\frac{1}{r}\left[\frac{1}{\sin\theta}\frac{\partial}{\partial\phi}(C_r)-\frac{\partial}{\partial r}\left(rC_\phi\right)\right]\hat{a}_\theta$$

$$+\frac{1}{r}\left[\frac{\partial}{\partial r}(rC_\theta)-\frac{\partial}{\partial\theta}(C_r)\right]\hat{a}_\phi$$

$$\nabla\times\vec{C}=\frac{1}{r\sin\theta}\left[\frac{\partial}{\partial\theta}(\cos\theta\sin\theta)-\frac{\partial}{\partial\phi}(r\sin\theta\cos\phi)\right]\hat{a}_r$$

$$+\frac{1}{r}\left[\frac{1}{\sin\theta}\frac{\partial}{\partial\phi}\left(\frac{\cos\theta}{r^2}\right)-\frac{\partial}{\partial r}(r\cos\theta)\right]\hat{a}_\theta$$

$$+\frac{1}{r}\left[\frac{\partial}{\partial r}\left(r^2\sin\theta\cos\phi\right)-\frac{\partial}{\partial\theta}\left(\frac{\cos\theta}{r^2}\right)\right]\hat{a}_\phi$$

$$\nabla\times\vec{C}=\frac{1}{r\sin\theta}(\cos 2\theta+r\sin\theta\sin\phi)\,\hat{a}_r$$

$$+\frac{1}{r}(0-\cos\theta)\,\hat{a}_\theta$$

$$+\frac{1}{r}\left(2r\sin\theta\cos\phi+\frac{\sin\theta}{r^2}\right)\hat{a}_\phi$$

$$\nabla\times\vec{C}=\left(\frac{\cos 2\theta}{r\sin\theta}+\sin\phi\right)\hat{a}_r-\frac{\cos\theta}{r}\,\hat{a}_\theta$$

$$+\left(2\cos\phi+\frac{1}{r^3}\right)\sin\theta\hat{a}_\phi$$

Rack your Brain

Is Stokes' theorem related to Helmholtz theorem?

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Laplacian operators can only work on a scalar fields.

Laplacian Operator

- Laplacian operator is the combination of gradient and divergence operators.
- In other words, it is described as the divergence of a gradient of scalar field V.
- Mathematically, it is represented as-

  Laplacian V = divergence of (gradient of V)

  $= \nabla \cdot (\nabla V)$

  Laplacian V = $\nabla^2 V$

Rack your Brain

Does laplacian operator has negative values also?

COORDINATE SYSTEM	LAPLACIAN OF SCALAR V
Cartesian Coordinates (x, y, z)	$\nabla^2 V = \frac{\partial^2 V}{\partial x^2} + \frac{\partial^2 V}{\partial y^2} + \frac{\partial^2 V}{\partial z^2}$
Cylindrical Coordinates (ρ, ϕ, z)	$\nabla^2 V = \frac{1}{\rho}\frac{\partial}{\partial \rho}\left(\rho \frac{\partial V}{\partial \rho}\right) + \frac{1}{\rho^2}\frac{\partial^2 V}{\partial \phi^2} + \frac{\partial^2 V}{\partial z^2}$
Spherical coordinates (r, θ, ϕ)	$\nabla^2 V = \frac{1}{r^2}\frac{\partial}{\partial r}\left(r^2 \frac{\partial V}{\partial r}\right) + \frac{1}{r^2 \sin\theta}\frac{\partial}{\partial \theta}\left(\sin\theta \frac{\partial V}{\partial \theta}\right) + \frac{1}{r^2 \sin^2\theta}\frac{\partial^2 V}{\partial \phi^2}$

Table 1.3 : Laplacian of scalar field V in different coordinate system

- The Laplacian of a scalar field gives another scalar field.
- A scalar field V will be considered as harmonic in a given region if its Laplacian is zero (i.e., it vanishes in that zero).
- Mathematically, $\nabla^2 V = 0$ (Laplacian equation) here harmonic V would be in the form of sine or cosine.
- Important property of Laplacian operator:

  $\nabla^2 \vec{A} = \nabla(\nabla \cdot \vec{A})$ and here in cartesian system,

$$\nabla^2 \vec{A} = \nabla^2 A_x \hat{a}_x + \nabla^2 A_y \hat{a}_y + \nabla^2 A_z \hat{a}_z$$

Previous Year's Questions

Analyze the Laplacian of the following scalar field:

$f(x, y, z) = (x^2 + y^2 + z^2)^{-1/2}$

Example 10:

Find the Laplacian of the given scalar fields.

(i) $P = x^2y + x^2yz - yz^3$

(ii) $Q = \rho z \sin\phi + z^2 \cos^2\phi + \rho^2$

(iii) $T = 2r\cos\theta\cos\phi$

Solution:

(i) In Cartesian coordinates system:

$$\nabla^2 P = \frac{\partial^2 P}{\partial x^2} + \frac{\partial^2 P}{\partial y^2} + \frac{\partial^2 P}{\partial z^2}$$

$$= \frac{\partial}{\partial x}(2xy + 2xyz) + \frac{\partial}{\partial y}\left(x^2 + x^2z - z^3\right)$$

$$+ \frac{\partial}{\partial z}\left(x^2y - 3yz^2\right)$$

$$= 2y + 2yz + 0 - 6yz$$

(ii) $Q = \rho\, z \sin\phi + z^2\cos^2\phi + \rho^2$

$$\nabla^2 Q = \frac{1}{\rho}\frac{\partial}{\partial\rho}\left(\rho\frac{\partial Q}{\partial\rho}\right) + \frac{1}{\rho^2}\frac{\partial^2 Q}{\partial\phi^2} + \frac{\partial^2 Q}{\partial z^2}$$

$$= \frac{1}{\rho}\frac{\partial}{\partial\rho}\left[\rho\left(z\sin\phi + 2\rho\right)\right]$$

$$+ \frac{1}{\rho^2}\frac{\partial}{\partial\phi}\left[\rho z\cos\phi - 2z^2\sin\phi\cos\phi\right]$$

$$+ \frac{\partial}{\partial z}\left[\rho\sin\phi + 2z\cos^2\phi\right]$$

$$= \frac{1}{\rho}\left[z\sin\phi + 4\rho\right] + \frac{1}{\rho^2}\left[-\rho z\sin\phi - z^2 2\cos 2\phi\right]$$

$$+ \left[0 + 2\cos^2\phi\right]$$

$$= \frac{z}{\rho}\sin\phi + 4 - \frac{z}{\rho}\sin\phi - \frac{2z^2}{\rho^2}\cos 2\phi + 2\cos^2\phi$$

$$= 4 + 2\cos^2\phi - \frac{2z^2}{\rho^2}\cos 2\phi$$

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Uses of Laplacian:

The Laplacian occurs in many differential equations that define several physical phenomena, like electric and gravitational potentials, fluid flow, wave propagation, and quantum mechanics.

Rack your Brain

The Laplacian operator is a linear operator- (True /False)?

(iii) $T = 2r\cos\theta\cos\phi$

$$\nabla^2 T = \frac{1}{r^2}\frac{\partial}{\partial r}\left[r^2\frac{\partial T}{\partial r}\right] + \frac{1}{r^2\sin\theta}\frac{\partial}{\partial\theta}\left[\sin\theta\frac{\partial T}{\partial\theta}\right]$$

$$+\frac{1}{r^2\sin^2\theta}\frac{\partial^2 T}{\partial\phi^2}$$

$$= \frac{1}{r^2}\frac{\partial}{\partial r}\left[r^2(2\cos\theta\cos\phi)\right]$$

$$+\frac{1}{r^2\sin\theta}\frac{\partial}{\partial\theta}\left[\sin\theta(-2r\cos\phi\sin\theta)\right]$$

$$+\frac{1}{r^2\sin^2\theta}\frac{\partial}{\partial\phi}(-2r\cos\theta\sin\phi)$$

$$= \frac{1}{r^2}(4r\cos\theta\cos\phi)$$

$$+\frac{1}{r^2\sin\theta}\left[(-2r\cos\phi\times 2\sin\theta\cos\theta)\right]$$

$$+\frac{1}{r^2\sin^2\theta}\left[-2r\cos\theta\cos\phi\right]$$

$$= \frac{4\cos\theta\cos\phi}{r} - \frac{4\cos\theta\cos\phi}{r} - \frac{2\cos\theta\cos\phi}{r^2\sin^2\theta}$$

$$= \frac{-2}{r}\cot\theta\ \mathrm{cosec}\,\theta\ \cos\phi$$

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In a cartesian coordinate system, the Laplacian is obtained by the sum of second partial derivatives of the function w.r.t every independent variable. but in cylindrical or spherical coordinates, the Laplacian also has a distinct form.

CHAPTER SUMMARY

RECTANGULAR TO CYLINDRICAL	CYLINDRICAL TO RECTANGULAR
Variable change $\begin{cases} x = \rho\cos\phi \\ y = \rho\sin\phi \\ z = z \end{cases}$	Variable change $\begin{cases} \rho = \sqrt{x^2+y^2} \\ \phi = \tan^{-1}\left(\frac{y}{x}\right) \\ z = z \end{cases}$ $\begin{cases} \sin\phi = \frac{y}{\sqrt{x^2+y^2}} \\ \cos\phi = \frac{x}{\sqrt{x^2+y^2}} \end{cases}$
Component change $\begin{cases} A_\rho = A_x\cos\phi + A_y\sin\phi \\ A_\phi = -A_x\sin\phi + A_y\cos\phi \\ A_z = A_z \end{cases}$	Component change $\begin{cases} A_x = A_\rho\frac{x}{\sqrt{x^2+y^2}} - A_\phi\frac{y}{\sqrt{x^2+y^2}} \\ A_y = A_\rho\frac{y}{\sqrt{x^2+y^2}} + A_\phi\frac{x}{\sqrt{x^2+y^2}} \\ A_z = A_z \end{cases}$

RECTANGULAR TO SPHERICAL	SPHERICAL TO RECTANGULAR
Variable change $\begin{cases} x = r\sin\theta\cos\phi \\ y = r\sin\theta\sin\phi \\ z = r\cos\theta \end{cases}$	Variable change $\begin{cases} r = \sqrt{x^2+y^2+z^2} \\ \theta = \tan^{-1}\left[\frac{\sqrt{x^2+y^2}}{z}\right] \\ \phi = \tan^{-1}\left(\frac{y}{x}\right) \end{cases}$ $\begin{cases} \cos\phi = \frac{x}{\sqrt{x^2+y^2}} \\ \sin\phi = \frac{y}{\sqrt{x^2+y^2}} \end{cases}$

1. The three common coordinate systems are the cartesian (or rectangular), the cylindrical and the spherical.

2. A point T is represented as T(x, y, z), T(ρ, ϕ, z), and T(r, θ, ϕ) in the cartesian, cylindrical, and spherical systems respectively.

3. Mathematical operations (addition, subtraction, product etc.) can be performed in the same coordinate system.

4. Fixing one space variable defines a surface; fixing two variables defines a line; fixing three variables defines a point.

5. A unit normal vector is $\hat{a}_n$ which is normal to surface where n denotes constant.

6. The differential length in the cartesian, cylindrical, and spherical systems are respectively,

$$d\vec{l} = dx\,\hat{a}_x + dy\,\hat{a}_y + dz\,\hat{a}_z$$
$$d\vec{l} = d\rho\,\hat{a}_\rho + \rho\,d\phi\,\hat{a}_\phi + dz\,\hat{a}_z$$
$$d\vec{l} = dr\,\hat{a}_r + r d\theta\,\hat{a}_\theta + r\sin\theta\,d\phi\,\hat{a}_\phi$$

Note that $d\vec{l}$ is always taken to be in the positive direction; the direction of the displacement is taken care of by the limits of integration.

7. The differential normal areas in the three coordinate systems are respectively,

$$\left.\begin{aligned} d\vec{S} &= dy\ dz\,\hat{a}_x \\ d\vec{S} &= dx\ dz\,\hat{a}_y \\ d\vec{S} &= dx\ dy\,\hat{a}_z \end{aligned}\right\} \text{Cartesian system}$$

$$\left.\begin{aligned} d\vec{S} &= \rho\,d\phi\,dz\,\hat{a}_\rho \\ d\vec{S} &= d\rho\,dz\,\hat{a}_\phi \\ d\vec{S} &= \rho\,d\rho\,d\phi\,\hat{a}_z \end{aligned}\right\} \text{Cylindrical system}$$

$$\left.\begin{aligned} d\vec{S} &= r^2\sin\theta\,d\theta\,d\phi\,\hat{a}_r \\ d\vec{S} &= r\sin\theta\,dr\,d\phi\,\hat{a}_\theta \\ d\vec{S} &= r\,dr\,d\theta\,\hat{a}_\phi \end{aligned}\right\} \text{Spherical system}$$

Note that $d\vec{S}$ can be in the positive or negative direction depending on the surface under consideration.

8. The differential volumes in the three coordinate systems are-

$$dV = dxdydz$$

$$dV = \rho \, d\rho \, d\phi \, dz$$

$$dV = r^2 \sin\theta dr \, d\theta \, d\phi$$

PRACTICE QUESTIONS

Q1 Find the divergence of the vector field $\vec{F}(\theta) = -\sin\theta\hat{i} + \cos\theta\hat{j}$

(A) $\nabla\cdot\vec{F} = \frac{\sin 2\theta}{r}$ (B) $\nabla\cdot\vec{F} = -\cos\theta - \sin\theta$

(C) $\nabla\cdot\vec{F} = 0$ (D) $\nabla\cdot\vec{F} = -\frac{2\sin\theta\cos\theta}{r}\hat{i} + \frac{2\sin\theta\cos\theta}{r}\hat{j}$

Q2 For the cylindrical coordinate system, the coordinate ρ-curves are-
(A) vertical straight lines.
(B) horizontal circles with centers on the z-axis.
(C) horizontal straight lines.
(D) horizontal straight half-lines radiating from the z-axis.

Q3 Find the gradient of the scalar field $f(r, \theta, z) = r^2\theta z$ in terms of the cylindrical coordinates.

(A) $\nabla f = 2r\theta z\hat{a}_r + r\hat{a}_\theta + r^2\theta\hat{a}_z$ (B) $\nabla f = 2r\theta z\hat{a}_r + rz\hat{a}_\theta + r^2\theta\hat{a}_z$

(C) $\nabla f = 2r\theta z\hat{a}_r + r\theta z\hat{a}_\theta + r^2\theta\hat{a}_z$ (D) $\nabla f = r\theta z\hat{a}_r + r\theta z\hat{a}_\theta + r^2\theta\hat{a}_z$

Q4 Use Stokes' Theorem to evaluate $\iint_s (\text{curl}\,\vec{F})\cdot d\vec{S}\,\hat{a}_n$ where $\vec{F} = x^2\hat{i} + 2xz^2\hat{j} + 3y^2z\hat{k}$ and S is the surface of the paraboloid $z = 4 - x^2 - y^2$.

(A) $4\pi a^2$ (B) $\frac{1}{2}\pi a^2$ (C) $a\pi^2$ (D) 32π

Q5 Use the Divergence Theorem to calculate the flux $\oiint_s \vec{F}\cdot\vec{N}\,dS$ of the vector field $\vec{F} = (x^2 + \sin(yz))\hat{i} + (y - xe^{-z})\hat{j} + (z^2)\hat{k}$ out of the surface S of the region bounded by the cylinder $x^2 + y^2 = 4$ and planes $x + z = 2$ and $z = 0$.

(A) 20π (B) $\frac{10\sqrt{2}}{3}$ (C) $10\sqrt{2}$ (D) $\frac{20\sqrt{2}}{3}$

Q6 Let S be an imaginary closed surface in an ideal fluid which bounds a domain D. The conservation of mass states that the rate of change of the mass of fluid in D equals the rate at which fluid enters D across S. Given that the mass changes at rate $\frac{\partial}{\partial t}\iiint_D \delta(x,y,z,t)\,dV$ (where $\delta(x, y, z, t)$ is the density) and the rate at which mass is flowing out D across S is $\oiint_S \delta(x,y,z,t)\,v(x,y,z,t)\cdot\hat{a}_n\,dS$, it follows that:

(A) $\frac{\partial \delta}{\partial t} + \text{div}(\delta v) = 0$

(B) $\frac{\partial \delta + \text{div}(\delta v)}{\partial t} = 0$

(C) $\frac{\partial \delta}{\partial t} - \text{div}(\delta v) = 0$

(D) $\text{div}(\delta v) = 0 \Leftrightarrow \text{div}(v) = 0$

Q7 Suppose a closed surface S bounds a region R in which charge is distributed with density ρ. Given that the electric field E due to the charge satisfies div E = $k\rho$, and the total charge in R is Q, find the total flux of E out of R through S.

(A) $\pi k^2 q$ (B) kQ (C) $k\rho Q$ (D) πkQ

Q8 Suppose the vector-valued function $\vec{r}(t)$ satisfies $\vec{r}'(t) = \left(-6t^2, 2t+1, 8t^2\right)$ and $\vec{r}(0) = (1, 2, 3)$. Find $\vec{r}(1)$

(A) $(-1, 4, 5)$ (B) $(-2, 2, 2)$ (C) $(-6, 3, 8)$ (D) $(-11, 3, 27)$

Q9 Which of the following are vector quantities?

i. The velocity of a frisbee.
ii. The width of a crater made by an asteroid.
iii. The speed of a car on the highway.
iv. The displacement of a billiard ball after it is struck by the cue ball.

(A) i only (B) i and ii (C) ii and iii (D) i and iv

Q10 Determine which of the following expression gives the length of the curve defined by $\vec{r}(t) = 2t\hat{i} + \cos t\hat{j} + 2\sin t\hat{k}$ between the point (0, 1, 0) and $(2\pi, -1, 0)$.

(A) $\int_0^{\pi} \sqrt{4 + \sin^2 t + 4\cos^2 t}\, dt$

(B) $\int_0^{\pi} \sqrt{4t^2 + \cos^2 t + 4\sin^2 t}\, dt$

(C) $\int_0^{2\pi} \sqrt{4t^2 + \cos^2 t + 4\sin^2 t}\, dt$

(D) $\int_0^{2\pi} \sqrt{4 + \sin^2 t + 4\cos^2 t}\, dt$

Answer key:

Q1	(C)	**Q2**	(D)	**Q3**	(B)	**Q4**	(D)	**Q5**	(A)
Q6	(A)	**Q7**	(B)	**Q8**	(A)	**Q9**	(D)	**Q10**	(A)